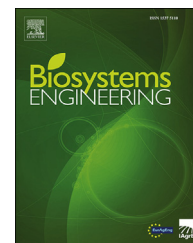


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Review

Intelligent fish feeding based on machine vision: A review



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Feeding is an important link in aquaculture that not only affects the healthy growth of fish but is also the main factor that determines the cost and economic benefits of aquaculture. At present, feeding in aquaculture is mainly based on mechanical timing and quantity, relying on a fixed amount of feed preset by the breeder, which does not take into account the dynamic feeding needs of the fish. Problems of underfeeding or overfeeding may occur, affecting the growth rate of the fish and polluting the aquaculture water. Machine vision is an efficient, economical, non-destructive, and objective detection and analysis technology, which is of great significance in promoting the automation and intelligence of aquaculture. Therefore, the combination of machine vision and fish feeding demand for feeding is helpful to improve efficiency and achieve intelligent aquaculture. This paper reviews the research on intelligent fish feeding based on machine vision. The main factors affecting feeding are introduced, followed by a general description of the process of intelligent fish feeding based on machine vision, and a detailed overview and analysis of the key technologies involved in image acquisition, image processing and analysis, and intelligent fish feeding. It also discusses the challenges and potential solutions in intelligent feeding. In short, it aims to help researchers and industry practitioners to better understand the state of the art of machine vision in intelligent fish feeding, and to assist in promoting accurate feeding and improving the efficiency of aquaculture.

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1. Introduction

With improved consumption levels and changes in lifestyles and eating habits, people are paying more attention to the nutritional balance of their daily diets, and demand for high-protein, low-fat foods is increasing. Fish is rich in high-quality protein, has high nutritional value, and is very popular with consumers. Aquaculture is the main source of fish for human consumption. It is estimated that by 2030, two-thirds of the world's fish will be supplied by aquaculture (Li & Du, 2021). With the vigorous development of a new round of scientific and technological revolution, modern information technology provides technical support for the development of aquaculture automation and intelligence. Accurate and intelligent feeding of aquaculture is a key link in improving the level of fisheries informatisation and promoting the high-quality development of fisheries.

Feed provides nutrients and energy for fish survival, and plays an important role in aquaculture, mainly including plant-based fresh feed, animal-based fresh feed such as earthworms, commercial compound feed, and homemade pellet feed (Hao, 2016). Data show that feed costs account for more than 60% of the total cost of aquaculture. Therefore, reducing feed costs can greatly improve the economic benefits of aquaculture. Feed costs depend on the type and amount of feed. The type of feed is generally determined by the species of fish being farmed. Proper control of the feed amount has become the main method to reduce costs and maximise benefits (Føre et al., 2016). The key to achieving precise feeding in aquaculture is to provide an appropriate amount of feed in a given period of time to meet the nutrient requirements for the specific growth rate of the cultured fish. However, machine-timed quantitative feeding is currently mainly used in aquaculture, which relies on a fixed amount of feed preset by the breeder, which may lead to problems of underfeeding or overfeeding due to the inability to adjust to the feeding needs of the fish automatically. Underfeeding will affect the growth rate of the fish, and overfeeding will not only lead to feed waste and increase the cost of aquaculture, but also pollute the aquaculture water, and harmful substances such as ammonia nitrogen and nitrate compounds produced by the remaining feed will endanger the growth of the fish. It can be said that feeding not only affects the healthy growth of fish but also determines the economic benefits of aquaculture to a greater extent (Hu, Zhao, et al., 2021; Wu et al., 2015).

Machine vision is an interdisciplinary field involving many technologies such as artificial intelligence, computer science, image processing, and pattern recognition. The technology mainly uses computers to simulate human visual functions, extract information from the images of objective things, process and understand, and use in actual detection, measurement and control, which can greatly improve production efficiency and automation. Recently, machine vision technology has been studied and applied in aquaculture size measurement, quality classification, disease detection, species identification, and behaviour analysis (Banerjee et al., 2021; Ju & Xue, 2020; Lopez-Marcano et al., 2021; Wu et al.,

2019). The application of machine vision to aquaculture feeding is the focus and hotspot of current research, which will help to improve production efficiency, reduce breeding costs, promote the development of aquaculture and management towards automation and intelligence, and realise intelligent aquaculture feeding.

This paper mainly reviews the research on fish intelligent feeding based on machine vision. The main factors influencing feeding are presented in Section 2. The architecture and description of intelligent fish feeding based on machine vision are given in Section 3. Sections 4, 5, and 6 provide an overview and analysis of the key technologies involved in image acquisition, image processing and analysis, and intelligent fish feeding respectively. The challenges and potential solutions in intelligent feeding are discussed in Section 7, and Section 8 concludes the paper.

2. Main factors affecting the feeding

Feeding control based on fish feed demand can effectively ensure accurate feeding in aquaculture, i.e., how much feed do fish need, which is important to reduce the cost and improve the efficiency of aquaculture. However, the feeding demand of fish is not only influenced by the characteristics of the fish themselves but also changes due to the environment, feed, etc. Therefore, this section examines the influence of intrinsic fish characteristics, complex environment, and feed factors on feeding, as shown in Fig. 1.

2.1. Intrinsic characteristics of fish

The factors of the fish themselves are the fundamental factors influencing feeding. If the mass of all the fish in the nursery or the condition of the fish themselves changes, their diet will also change.

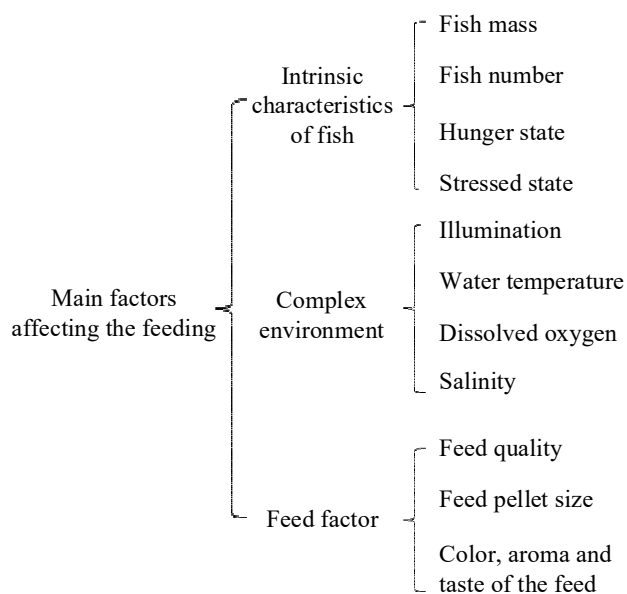


Fig. 1 – Main factors affecting feeding.

(1) Fish mass

Fish mass is the weight of the fish, which is a physiological attribute. Its change will affect the metabolism and feed intake of the fish. Generally speaking, the larger the fish mass, the larger its feed intake, i.e. the feed intake per unit mass decreases, that is, the feeding rate decreases. For example, the feeding rate of carp at 30–50g is 6.8% and is only 2.2% at 800 g.

(2) Fish number

Fish number is the total number of all fish in the breeding area. The more fish in the breeding area, the greater the feed demand and the more the feeding amount put in. The mass of all fish in the breeding area can be calculated from the fish number and the average fish mass.

(3) Hunger state

Fish exhibit a rapid burst of growth after re-feeding following starvation or nutritional deficiency, known as compensatory growth. Their feeding rate increases initially and then gradually decreases until it is constant. Studies have shown that, within certain limits, the longer a fish is starved, the higher the feeding rate during the satiation phase.

(4) Stressed state

When a fish is under stress, its eating rate will decrease, and the degree of decrease will depend on the species. For example, rainbow trout will usually fast for 4–7 days after a short transport and brown trout will stop eating for 3 days after a 2-min capture treatment. For domesticated ornamental fish such as goldfish, the stress response to a general capture treatment is weaker and the influence of feeding is small.

2.2. Complex environment

As fish live in a specific aquatic environment, their growth conditions are strongly influenced by the complex environment. The feeding rate will also change greatly when environmental factors such as light, water temperature, dissolved oxygen, and salinity change.

(1) Illumination

For fish that rely on visual feeding, light is one of the most important factors affecting the feeding rate. On the one hand, light can increase the contrast between the feed and the background, making it easier for fish to find, and on the other hand, light is closely related to the activity of feeding organisms. If the feed is plankton and larvae, they tend to congregate in the upper layer of the water in the morning and evening, which is more favourable for fish to feed on. Research shows that an appropriate range of light intensity is required for visual feeding by fish when fish are most active and feeding rates are highest.

(2) Water temperature

Water temperature also has an important impact on fish feeding rate. Generally speaking, within the appropriate temperature range, the higher the water temperature, the higher the feeding rate. Water that is too hot or too cold will reduce the feeding rate. However, different fish have different temperature ranges. For example, tilapia, a tropical fish, feeds the most and grows the fastest when the water temperature is around 30 °C. It grows normally at 25–35 °C and can't survive below 13–17 °C. Silver carp start feeding when the water temperature is above 8–10 °C, and when the water temperature is above 16–18 °C, the food intake will increase sharply and the growth will accelerate.

(3) Dissolved oxygen

Dissolved oxygen affects the oxygen uptake rate, and the uptake rate decreases when the dissolved oxygen level is low. In general, when dissolved oxygen saturation falls below 50% ~70%, the appetite of most fish decreases and so does the feeding rate. Fish with schooling habits, fast swimmers, and omnivores tend to have higher oxygen requirements.

(4) Salinity

Salinity also affects the feeding habits of some fish. Studies have shown that the optimum salinity for *Pagrosomus major* is 2.5%–3.0%. When the salinity is lower than 1.0%, the *Pagrosomus major* hardly feeds, and when the salinity is higher than 3.0%, the feeding rate decreases sharply. Too high or too low salinity will reduce the fish feeding rate of the fish.

In addition, if the water quality is polluted by the increased chemicals such as ammonia nitrogen, and nitrate in the water, the fish will feed less.

2.3. Feed factor

The nutrient composition, particle size, and hardness of the feed are also directly related to fish intake. In general, the higher the quality of feed, the lower the feeding rate, and vice versa. In particular, the protein and fat content of the feed will have the greatest effect on the amount fed. When feeding, it is also necessary to consider whether the particle size, hardness, and texture of the feed are suitable for the fish and whether other factors such as colour, aroma, and taste have a certain appeal to the fish, to ensure that the feed is a favourite food of the fish. The amount of feeding that is preferred is high, and the amount of feeding that is not preferred should be reduced accordingly.

3. Architecture of intelligent fish feeding based on machine vision

The key to achieving intelligent feeding in aquaculture is to provide the right amount of feed in a given period of time to meet the nutritional requirements for a given growth rate of the farmed fish. Fish mass, known as a physiological trait, is

the body weight of the fish. It is an intrinsic characteristic factor influencing fish feeding, which generally changes a little in a short period of time. The mass of all fish in the hatchery is calculated from the fish mass and the number of fish. The feeding amount determined based on the mass of all fish is the basis for achieving accurate fish feeding and avoiding unlimited feeding. The factors that influence fish feeding, such as hunger, stress, and complex environments, are constantly changing in a short period of time, and keeping track of changes in these factors can provide a more accurate picture of fish feeding requirements. In addition, the palatability of the feed also affects fish feeding requirements to some extent. Therefore, the key to achieving accurate fish feeding is to analyse the changes in these feeding factors, keep track of the fish feeding demand and adjust the feeding amount. However, there may be a coupling between several feeding factors, and the dynamic fusion analysis of them is complex and difficult to achieve.

Feeding behaviour is the movement of fish to feed. When the physiological status, nutritional requirements, and environmental conditions of fish change, it is directly reflected in their behaviour (Bao et al., 2019; Doyle et al., 2016; Yang, Zhang, et al., 2021). It can be argued that fish feeding behaviour may reflect changes in the factors influencing fish feeding. In addition, feed fed into the water is consumed by fish. It indicates a weakened feeding demand by fish and also reflects changes in the factors influencing fish feeding when there are more feed residues in the water column per unit of time. Therefore, the analysis of fish feeding behaviour and the detection of feed in aquaculture water can both reflect the changes in the factors influencing fish feeding, so that the fish feeding demand can be obtained and intelligent feeding decisions can be made. Figure 2 shows the process of intelligent

fish feeding based on machine vision as presented in the paper.

Intelligent fish feeding based on machine vision consists of three parts: image acquisition, image processing and analysis, and intelligent feeding decision. In image acquisition, data such as fish body images, fish school images, and fish feeding behaviour videos are acquired by the image acquisition equipment. In image processing and analysis, operations of pre-processing, feature extraction, and classification regression are performed on the acquired data to achieve fish mass estimation, fish counting, feeding behaviour analysis, aquaculture water feed detection, etc. In the feeding decision, the mass of all fish in the culture area is obtained based on the fish mass and fish number, then the feeding amount is preliminarily determined. In the feeding process, the feeding amount is adjusted in time based on the fish feeding demand obtained from the results of fish feeding behaviour analysis and aquaculture water feed detection to realise intelligent fish feeding.

4. Image acquisition

The processing and modelling of different fish images enables the automatic fish identification and the fish counting, which is the basis for intelligent fish feeding in aquaculture. However, the building of different models requires the support of fish image data with different semantics. Training with a large amount of data is the only way to ensure that the constructed models are well generalised. In addition, the quality of the fish images used will also influence the complexity of the image processing and determine the effectiveness of the analysis. Therefore, the acquisition of fish images is of particular

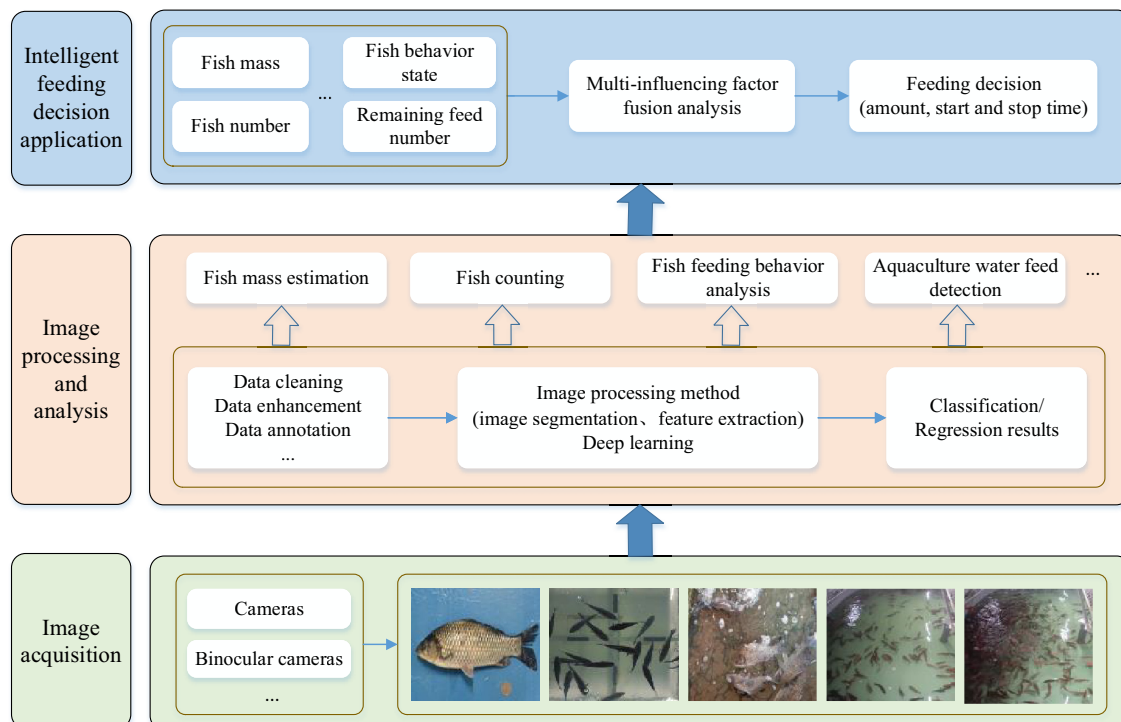


Fig. 2 – Aquaculture intelligent feeding method process.

importance. Nowadays, there are three main sources of fish image data: public data, Internet data, and experimental collection data.

4.1. Public data

Public data refers to the large-scale data collection used by various companies and organisations for machine learning, deep learning, and artificial intelligence research. The public data of fish images mainly include Fish 4 Knowledge (Qin et al., 2015), LifeCLEF (Salman et al., 2016), QUT fish databases (Sarigül & Avci, 2017), Kaggle4 (Ali-Gombe et al., 2017), and WHOI-Plankton (Lee et al., 2016). This type of data has the advantage of being large and easily accessible. However, the data come from different environments, resulting in uneven quality and unbalanced types, which increases the difficulty of fish image analysis. In addition, the fish images provided by public data can only be analysed for a single type of information. Most of the public data can only be used for studies such as fish species identification and fish target identification, but may not for studies such as fish mass estimation and fish feeding behaviour analysis. On the other hand, the fish images provided by public data may be images of a particular stage of several fish species, which cannot reflect the continuity of the fish growth process and cannot achieve the monitoring and analysis of the whole process of fish growth.

4.2. Internet data

Internet data refers to the data searched on the Internet, which has the advantages of large volume, convenient retrieval, and rich variety, but has the problem of irrelevance of some data. It is necessary to manually eliminate these irrelevant Internet-searched data when constructing the dataset. In addition, Internet data come from a variety of sources. Different collection personnel, locations, and equipment will result in inconsistent image quality and size. If no manual screening is not carried out, it will also increase the difficulty of analysis to some extent and affect the analysis effect. There are many fish images on the Internet, but they are very heterogeneous, and many of them are images of non-real fish in non-real scenarios, such as drawn fish cartoon images, fish images in virtual scenarios, etc. These images can't reflect the real growth conditions of fish and their role in realising intelligent feeding in aquaculture is limited.

4.3. Experimental collection data

Experimental data collection means that researchers select experimental bases, culture environments, and objects, build experimental platforms and acquire image data themselves. In this method, researchers can choose the specific research object, shooting angle, and shooting method according to their needs, and have some control over the images. Generally, the experimental acquisition methods include single-angle and multi-angle. Single-angle method refers to the fish images collected from only one angle such as the top view or side view. The experimental platform of this data acquisition method is simple to build and the data processing workload is low. However, it can only capture the changes in fish

characteristics in a certain projection plane and cannot fully describe the changes in fish body morphology and behaviour, so the accuracy of fish analysis needs to be improved. The multi-angle method refers to the fish images collected from multiple angles such as the top and the side of the fish target, as well as underwater. Compared with the single-angle method, it requires multiple acquisition devices to be installed at different locations to obtain fish information from different perspectives, so the construction of the experimental platform is more complicated. In addition, it is difficult to map multiple images acquired from different angles at the same time to discover the relationships between them and to construct a complete three-dimensional spatial model of the fish body. However, the construction of a three-dimensional spatial model can more comprehensively describe the changes in the shape and behaviour of the fish body and help to improve the accuracy of fish analysis. Therefore, single-angle method is suitable for scenarios with low requirements for analysis accuracy and no need to dynamically analyse the comprehensive morphology and behaviour of fish, while multi-angle method is suitable for scenarios with high requirements for analysis accuracy. A schematic diagram of a multi-angle image acquisition platform with top, side, and underwater views is shown in Fig. 3.

An experimental acquisition platform, usually consisting of a breeding pond, breeding objects, cameras, computers, and aerators. The camera is fixed at a predetermined angle with a bracket and video data is recorded from the cultured objects in the breeding pond. The camera and the computer are connected by a network cable and the collected multi-angle video data is transferred to the computer for storage. There are many frames in 1 s of the video, and each frame is an image. The collected video data is converted into image data by the code to construct the fish image dataset. Figure 4 shows the fish image data collected experimentally.

Fig. 4(a), (b), (c), (d), and (e) are fish images taken in the transport box of the aquaculture base, the laboratory platform, the sales pond of the aquatic product market, the aquaculture pond in the industrial workshop, and the pool, respectively. Figure 4(b) is the fish image taken by the camera installed directly above the fish pond so that the image size of different fish is the same. Fig. 4(a), (c), (d), and (e) are all taken from the top view, but the image acquisition device is at different angles and distances. The fish body closer to the imaging device in the image is larger and vice versa. Fish

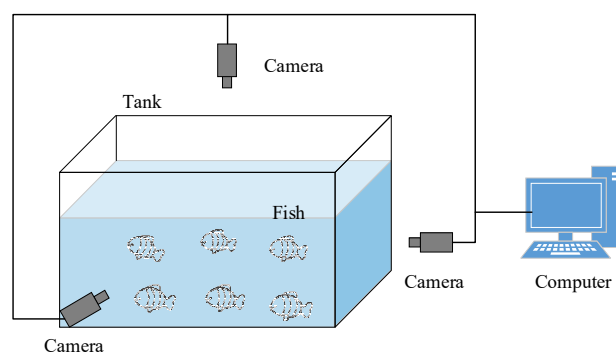


Fig. 3 – Schematic diagram of the multi-angle image acquisition platform with top, side, and underwater views.

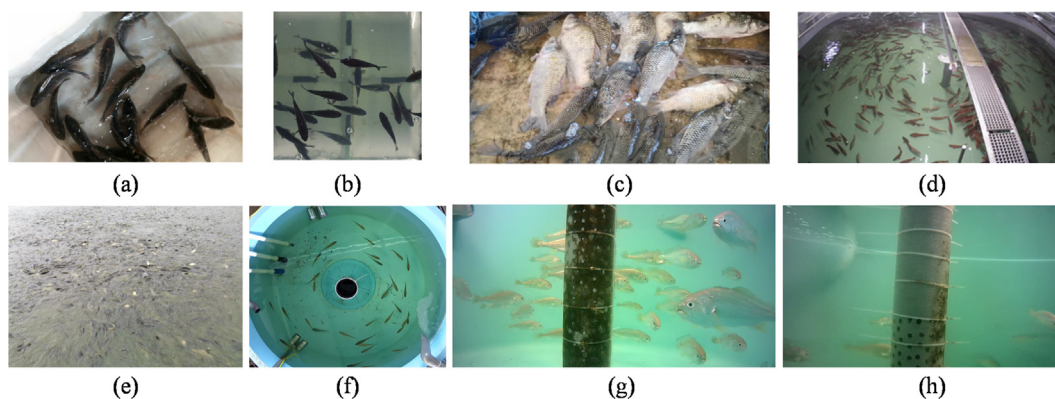


Fig. 4 – Fish image data.

images taken in different environments and using different imaging methods have different backgrounds, image sizes, and image lighting. In addition, the density of fish schools is different in different images. For example, there are few fish in Fig. 4(a) and (b), while Fig. 4(e) has a lot of fish collected from a high-density farming site, and in Fig. 4(d), the density of fish in different areas is uneven and there are many obstructions in the image. All these characteristics make fish image analysis face challenging.

Fig. 4(f), (g), and (h) are the fish images collected from different angles under the same platform, which are a top view, side view, and underwater view, respectively. Images from different angles reflect different information about the fish. As shown in Fig. 4(f), the top-view image shows the back and movement information of the fish. The side view image in Fig. 4(g) shows the abdominal information of the fish as well as its size and movement information. Figure 4(h) is an image obtained by installing the image imaging device in the water. The number of fish in it is less than in the top and side view images. The fish bodies in Fig. 4(g) and (h) are both imaged at different sizes. The integration of multi-angle data for joint analysis is helpful for a more comprehensive understanding of the fish size, behaviour, and other information.

Furthermore, binocular cameras are a special case in terms of multi-angle information acquisition. A binocular camera

consists of two cameras that observe the same object from two different viewpoints to obtain perceptual images under different viewpoints. Information about the size of an object is then obtained by calculating the positional deviation between image pixels using the imaging ensemble principle. Compared with ordinary cameras, which can only obtain two-dimensional images of the target, binocular cameras acquire multiple images of the object and reconstruct the three-dimensional information of the object through binocular image calibration, stereo correction, and stereo matching to realise the extraction of three-dimensional spatial features of the object. Currently, there are some preliminary studies and applications of binocular vision technology in fish size estimation and fish mass estimation in aquaculture. For example, in the study of fish mass estimation, since the mass of a fish body is not only related to its two-dimensional plane features such as length, width, and area, but also the thickness of the fish body has a great influence on the mass. Therefore, the three-dimensional information of the fish body is obtained by the binocular camera, and the thickness features of the fish body are extracted by combining with the detection of the key feature points of the fish body, which can be added to the quality prediction to further improve the estimation accuracy. The extraction process of the three-dimensional spatial features of the fish body is shown in Fig. 5.

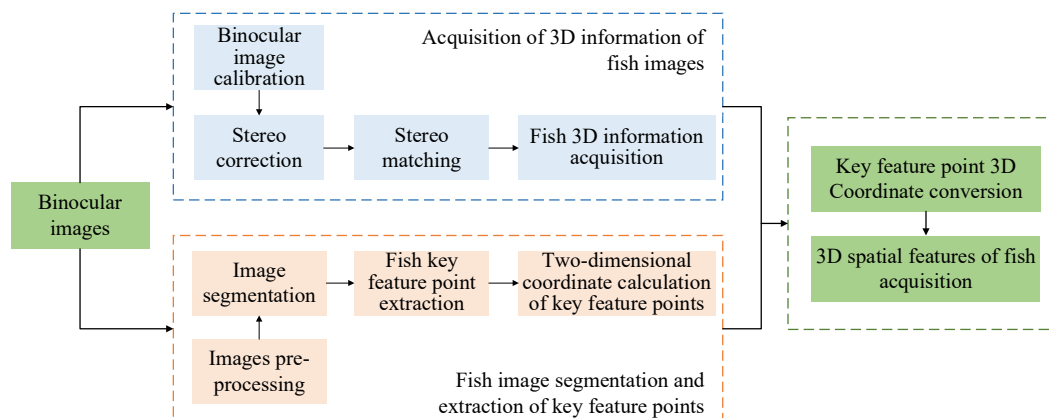


Fig. 5 – The extraction process of three-dimensional spatial features of fish body.

5. Image processing and analysis

Image processing refers to the removal of irrelevant information from the image and the extraction of the required key image information. It is the basis of image analysis, which mainly includes operations such as image denoising, image enhancement, image segmentation, image binarisation, morphological processing, and contour extraction. Image analysis includes operations such as image regression, image classification, and image clustering. Through image analysis, information about the target object can be obtained so that relevant decisions can be made. In intelligent fish feeding, regression analysis of fish images can be used to obtain information on physiological characteristics such as fish mass and fish number in the breeding area, and classification and recognition analysis of fish images can be used to obtain fish feeding states and changes in fish feeding demand, etc. Based on these analysis results, a basis for intelligent feeding decisions can be provided. The analysis results can provide the basis for intelligent feeding decisions. Depending on the image processing and analysis techniques used, it can be divided into traditional image processing and analysis method, and deep learning.

5.1. Traditional image processing and analysis

Traditional image processing and analysis is a feature-based method in which image processing is the first step, followed by image analysis. During image processing, the image segmentation algorithm is usually used to separate the target object from the background. Of course, to ensure the segmentation effect of the target object and to improve the accuracy of the subsequent image analysis, the image is also usually denoised and morphologically processed. In the image analysis part, it mainly includes image feature extraction and target classification and regression. During the growth of fish, their appearance changes, such as size, shape, and colour features, and during the feeding process, the different feeding states lead to changes in fish texture and other features. Therefore, accurate extraction of features such as shape, texture, and colour of the target object is key to subsequent image analysis using machine learning algorithms. Based on the extracted features, feature fitting is carried out by machine learning algorithms to achieve classification or regression of the

image through model building, training, and testing to obtain information about the target object. The process of traditional image processing and analysis method is shown in Fig. 6.

In this type of method, accurate segmentation is the basis for subsequent feature extraction and will affect the effectiveness of target recognition and regression. In addition, the quality of feature extraction will directly affect the effectiveness of image classification and regression and thus image analysis. This section therefore introduces image segmentation, image feature extraction, image classification and regression in the context of intelligent fish feeding.

5.1.1. Image segmentation

Image segmentation is to separate the target from the background in an image. In fish mass estimation, accurate segmentation of the target fish has a great influence on the accuracy of the mass estimation, and fish counting is also affected by inaccurate segmentation. Therefore, accurate segmentation of the target object is the basis for subsequent image analysis. According to different segmentation principles, common image segmentation methods in aquaculture include the threshold-based method, edge detection-based method, and specific theory-based method. Among the specific theory methods, wavelet analysis, genetic algorithm, and active contour are commonly used. However, due to the complexity of the aquaculture environment, fish images have some problems, such as uneven light intensity, water mist, and blurred fish edges, making it common to incorporate multiple segmentation algorithms when actually performing fish image segmentation. Table 1 lists some applications of image segmentation technology in aquaculture.

However, high-density farming is usually adopted in actual aquaculture, resulting in a lot of overlap in the fish school image of the culture area. If the overlapping targets are not segmented, the fish counting results will produce large errors. Therefore, after the fish target is extracted from the background by the image segmentation technology, it is necessary to further segment the overlapping fish target, as shown in Fig. 7.

In the research of overlapping target segmentation, there are currently three types of methods: the watershed method (Duan et al., 2019; Song et al., 2014; Zhang et al., 2016), the ellipse fitting method (Gonzalez-Hidalgo et al., 2015; Lu et al., 2016), and the segmentation point detection method (Zhang, Li, et al., 2022; Zhang & Li, 2017). In previous research, a method based on point detection and pairing was proposed.

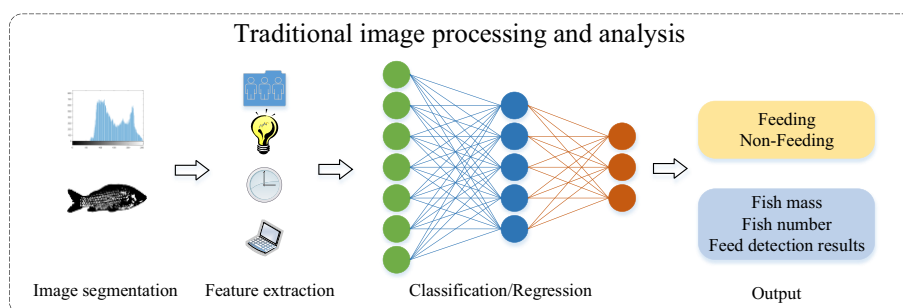


Fig. 6 – Traditional image processing and analysis process.

Table 1 – Applications of image segmentation in aquaculture.

Reference	Segmentation algorithm	Research object	Research objective
Al-Kofahi et al. (2013)	Two-dimensional wavelet	Fish egg nucleus	Count fish eggs
Hu et al. (2016)	Threshold method	Fish body	Estimate fish mass and guide feeding
Huang et al. (2016)	Gaussian mixture model	Fish body	Measure fish midline
Le and Xu (2016)	Improved adaptive threshold method	Fish body	Count fish
Ma et al. (2016)	Adaptive threshold method	<i>Cynoglossus gracilis</i>	Estimate fish mass
Williams et al. (2016)	Background difference method	Fish body	Estimate fish length
Issac et al. (2017)	Adaptive threshold method, and active contour method	Fish gill	Identify fish freshness
Malik et al. (2017)	Multiple edge detection algorithm	Fish body	Identify fish diseases
Qiao et al. (2017)	Canny edge detection algorithm, and C–V active contour method	Sea cucumber	Locate and identify sea cucumbers
Arora et al. (2018)	K-means clustering	Fish gill	Identify fish freshness
Saberloon and Cisar (2018)	Manually setting threshold method, and background difference method	Fish body	Estimate fish mass
Islamadina and Pramita (2018)	Canny edge detection algorithm	Tuna	Measure fish size
Azarmdel et al. (2019)	Manually setting threshold method	Head and belly of trout	Realise the loading and cleaning of fish meat
Duan et al. (2019)	Adaptive threshold method, and watershed algorithm	Fish eggs	Count fish eggs
Issac et al. (2019)	Adaptive threshold method	Fish body	Identify healthy and contaminated fish
Maurya et al. (2019)	Discrete wavelet	Fish gill	Identify normal and contaminated fish
Fan et al. (2021)	Improved Otsu threshold-segmentation method	Aquatic plants	Estimate the aquatic-plant area on a pond surface
Banwari et al. (2022)	Circle detection	Eyes of Roho Labeo	Estimate fish freshness
Hao et al. (2022)	C–V active contour method	Crucian carp	Estimate fish mass

The morphological factors were used to recognise the overlapping fish target, the segmentation points were searched by a method combining convex point detection and the shortest centroid distance, and the segmentation lines were determined based on the method of corner detection and non-maximum distance to segment the overlapping fish. The proposed method realised the automatic determination of the number of segmentation lines based on the overlapping shape of fish and ensured correct segmentation, which laid the foundation for accurate fish counting (Zhang, Li, et al., 2022).

5.1.2. Image feature extraction

Feature extraction is to find the representative information to represent the whole image. It is important for the accurate implementation of fish mass estimation, fish counting, feeding behaviour analysis, and feed detection in intelligent fish feeding. The shape feature, texture feature, space feature, and colour feature are the main common features involved in intelligent fish feeding.

(1) Shape features

Shape features are used to describe the shape of objects. Fish mass is closely related to shape features (Zion, 2012). Accurate extraction of fish shape features is helpful for effective fish mass estimation. In the fish mass estimation study by Zhang, Wang, and Duan (2020), the shape features related to fish mass estimation, such as length, width, and area, were extracted. To illustrate the meaning of each shape feature, the relevant feature variables are marked in a real fish body, as shown in Fig. 8.

Based on the feature variables shown in Fig. 8, the corresponding description and formula definition of each fish image feature extracted are given in Table 2.

The expression of shape features must be based on image processing and image segmentation, and its accuracy is affected by the segmentation effect. For poorly segmented images, the shape parameters cannot even be extracted.

(2) Texture features

Texture features describe the surface properties of objects that correspond to an image or image area (Chen et al., 2017). In fish feeding, many fish swim quickly to the feeding area to grab food. The direction of the fish's movement is chaotic, causing ripples and reflections on the water surface, resulting in corresponding changes in image texture. Therefore, the texture features of the fish image are very important for identifying the feeding status of the fish. The most commonly used statistical method for texture features is the grey-level co-occurrence matrix, including four key features energy, entropy, contrast, and correlation.

(3) Space features

Space features refer to the mutual spatial position or relative directional relationship between multiple objects segmented in an image (Zhang, Li, et al., 2020). In the fish counting study, due to the limitation of the camera view, fish at the edge of the field of view may be incompletely

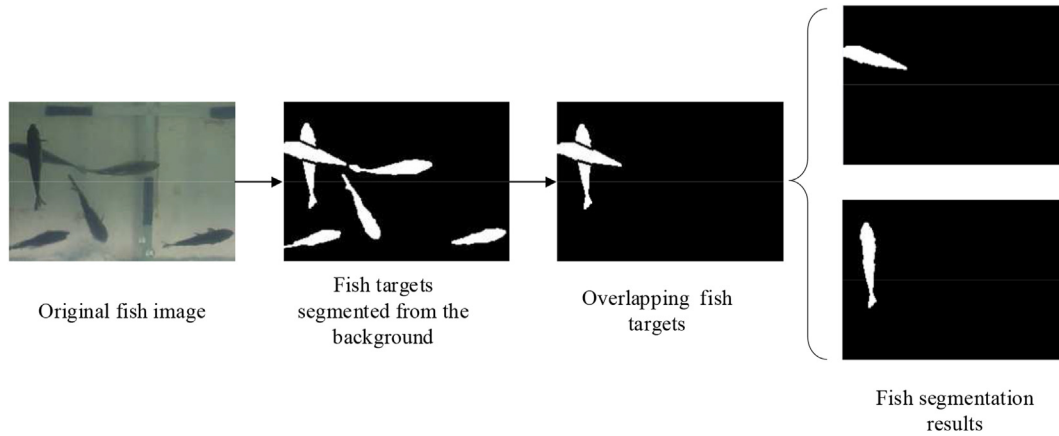


Fig. 7 – Overlapping fish image segmentation process.

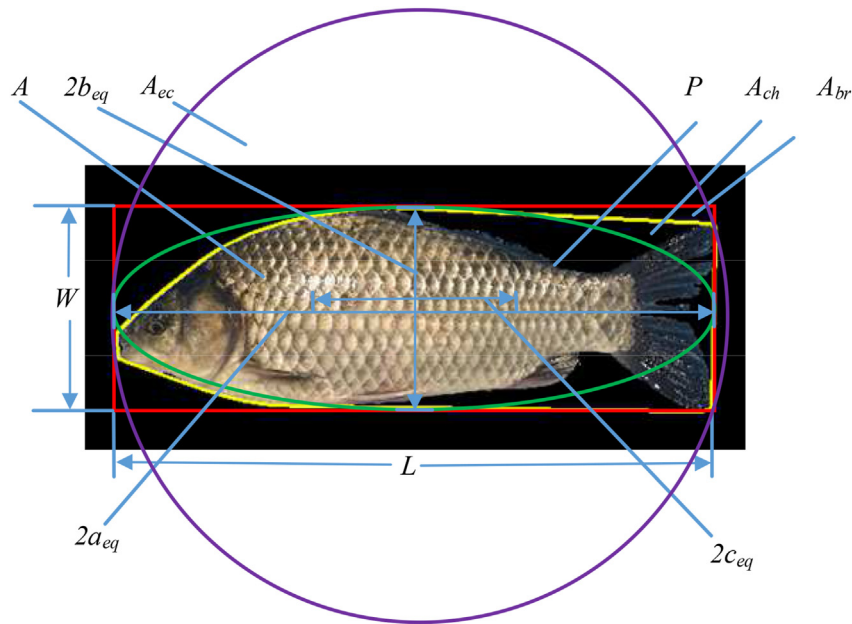


Fig. 8 – Illustration of fish feature variables.

Note: In Fig. 8, the figure constituted by red lines is the external rectangle of the fish contour, the figure constituted by green lines is the equivalent ellipse of the fish contour, the figure constituted by yellow lines is the convex hull of the fish contour, and the figure constituted by the purple lines is the circumcircle of the fish contour. A is the area of the fish, P is the perimeter of the fish contour, L and W are the length and width of the fish respectively. $2a_{eq}$, $2b_{eq}$, and $2c_{eq}$ are the long axis, short axis, and focal length of the equivalent ellipse of the fish contour, respectively. A_{ec} , A_{ch} , and A_{br} are the area of the circumcircle, the area of the convex hull, and the area of the external rectangle, respectively.

represented in the image, resulting in large differences in the relevant feature values in the connected areas with the same number of fish, as shown in the connected areas 1 and 2 in Fig. 9. In this case, the location of multiple connected areas in the image can be clarified by spatial relationship features, and fish can be counted more accurately.

Space features can improve the ability to distinguish image content, but they are often sensitive to rotation, inversion, and scale changes of images or objects, so they are usually used in conjunction with other features.

(4) Colour features

Colour feature is a global surface feature that can be visualised. This type of feature is obtained by converting pixel values into numerical values, including colour histograms, colour moments, etc (Mohammadi Lalabadi et al., 2020). The colour of the appearance of different species may be different. Therefore, it is important to combine colour features for fish species identification to determine the feeding rate of different fish species, and to obtain accurate estimates of the feeding amount.

Table 2 – Common fish shape features.

Feature name	Description of feature	Symbols and formulas
Area	Number of pixels in the fish region	A
Perimeter	Number of pixels on the boundary of the fish region	P
Length	The length of the fish	L
Width	The width of the fish	W
Equivalent diameter	The diameter of a circle equal to the area of the fish	D
Aspect ratio	Ratio of the length to the width of fish	$Ar = L/W$
Ration equivalent ellipse axis	Ratio of the long axis to the short axis of the fish equivalent ellipse	$Re = aeq/beq$
Eccentricity	Ratio of the distance between the focal points and the long axis of the fish equivalent ellipse	$Ec = ceq/aeq$
Compactness	Ratio of the equivalent circle diameter to the length	$Ca = D/L$
Heywood circularity factor	Description of the degree to which the target is closer to the circle	$Hcf = P / (2 * \sqrt{\pi * A})$
Extent	Ratio of the fish area to the external rectangular area	$Ex = A/Abr$
Solidity	Ratio of the fish area to the convex hull area	$So = A/Ach$
Arc	Ratio of the fish area to the circumcircle area	$Arc = A/Aec$
Complexity	Ratio of the square of the fish perimeter to the fish area	$Cl = P^2/A$

5.1.3. Image classification and regression

Based on the extracted fish image features, machine learning algorithms are used for feature fitting, image classification, and regression are used to perform operations such as fish behaviour state analysis and fish mass analysis. Fish feeding behaviour state identification is achieved by classification, while fish mass estimation and fish counting are achieved by regression. Back propagation neural network (BPNN) and support vector machine (SVM) can both perform classification and regression tasks and achieve good results with small sample data, making them the two most commonly used machine-learning algorithms in the implementation of intelligent fish feeding (Saberioon & Císar, 2018; Zhang, Wang, & Duan, 2020). Furthermore, regression models between one or two features and mass are constructed in many studies of fish mass estimation, the mapping relationship between features and results is not particularly complex, and better results can be obtained by linear fitting and polynomial fitting only, thus these two algorithms became the commonly used machine learning algorithms in this research.

5.2. Deep learning

Data is fed directly into the network and automatic feature extraction and combination is performed, and a large amount of labelled data is used for model training and validation to achieve image recognition and analysis, as shown in Fig. 10.

**Fig. 9 – Fish-connected area image.**

When deep learning is used to solve the problem of intelligent fish feeding, it mainly involves classification and detection algorithms, such as the implementation of fish feeding state identification and aquaculture water feed detection. MobileNet series algorithms are a kind of light-weight neural network classification algorithm with fewer network parameters and high speed (Howard et al., 2017; Howard et al., 2019; Sandler et al., 2018). The core of this series of algorithms is the depthwise separable convolution operation, which greatly reduces the number of parameters and can be applied to mobile devices, and thus is often used in fish feeding state identification. R-CNN series (Girshick, 2015; Girshick et al., 2014; Ren et al., 2017) and YOLO series (Bochkovskiy et al., 2020; Redmon et al., 2016; Redmon and Farhadi, 2017, 2018) are two types of common target detection algorithms in aquaculture. The former is a classical two-stage target detection algorithm with the advantage of accuracy, while the latter belongs to a one-stage target detection algorithm with the advantage of model size and detection speed. Both of them can achieve good results for the research of aquaculture water feed detection.

Deep learning can automatically extract low-level and high-level features from big data, identify subtle fish features from images, and is robust to changes in illumination, translation, and rotation, making it very suitable for solving classification and regression problems in aquaculture. However, deep learning requires a large number of samples for training, which is not available in some real scenes. Therefore, in the research of intelligent fish feeding in aquaculture, traditional image processing or deep learning should be selectively applied according to specific scenarios, research objects, and research objectives to ensure the analysis effect and promote the maximisation of aquaculture efficiency and benefits.

6. Key technologies for intelligent fish feeding

The information on fish mass, fish number, fish feeding behaviour, and feed in the aquaculture water was obtained based on the processing and analysis of the collected fish data. And then the intelligent feeding decision is made by

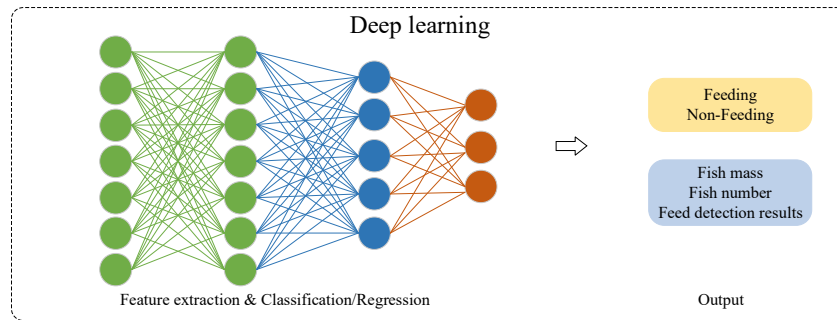


Fig. 10 – Deep learning process.

information fusion. Therefore, this section summarises the research progress of machine vision in intelligent fish feeding from the aspects of fish mass estimation, fish counting, fish feeding behaviour analysis, aquaculture water feed detection, and intelligent feeding decision.

6.1. Fish mass estimation

The mass of all fish in the culture area can be determined by studying the fish mass estimation, and the feeding rate of the cultured fish can be determined based on the fish mass and species, both of which are the basis for determining the feeding amount. During the breeding process, fish at different growth stages have different masses, as well as different feed requirements. By comparing the actual fish mass with the mass that the fish should have for normal healthy growth at a certain growth stage, it can be judged whether the growth condition of the fish is healthy and whether the feeding amount is reasonable, and then adjust the feeding and breeding mode in turn. Therefore, research on fish mass estimation is an important link in realising intelligent feeding and healthy breeding of fish.

The fish mass estimation method based on machine vision predicts the mass based on the body shape characteristics of fish, effectively avoids the problem of damage to the body surface of fish in the manual sampling and weighing method and has high efficiency and low damage. In this method, the image processing technology is used to preprocess the fish image to separate the fish from the background, fish image features are extracted, and feature values are calculated, and the relationship model between the feature values and the

mass is constructed to realise the fish mass estimation. The process is shown in Fig. 11. According to the number of extracted features, it can be divided into the single-factor method and the multiple-factor method (Duan, Li, et al., 2015; Zhang, Wang, & Duan, 2020).

6.1.1. Fish mass estimation method based on a single factor

The single-factor fish mass estimation method explores the relationship between fish mass and single image feature, establishes a relationship model between single feature and mass, and then realises fish mass estimation. Length and area are the two most common feature factors. In this type of method, the accuracy of fish mass estimation often depends on estimating the relationship between a single feature factor and mass.

Length is usually measured as the length of the line connecting the head to the tip of the tail of the fish. Currently, scholars have studied the relationship between length and mass (Froese et al., 2011; Venerus et al., 2016). The most classical length-mass relationship model was proposed by Fulton (1904) (Fulton, 1904) and the formula was $W = a \cdot L^b$, where W was the fish mass, L was the length, and a and b were the parameters empirically characterised fish species and strain parameters, respectively. The mass estimation method based on the length feature usually measures the length of the fish first and then estimates the fish mass through the length-mass relationship. Therefore, the accurate measurement of length becomes the key to these methods (Cheng et al., 2020; Garcia et al., 2020; Li et al., 2020; Muñoz-Benavent et al., 2018). Miranda and Romero (2017) collected images of the swimming rainbow trout in the water and measured the imaged length of

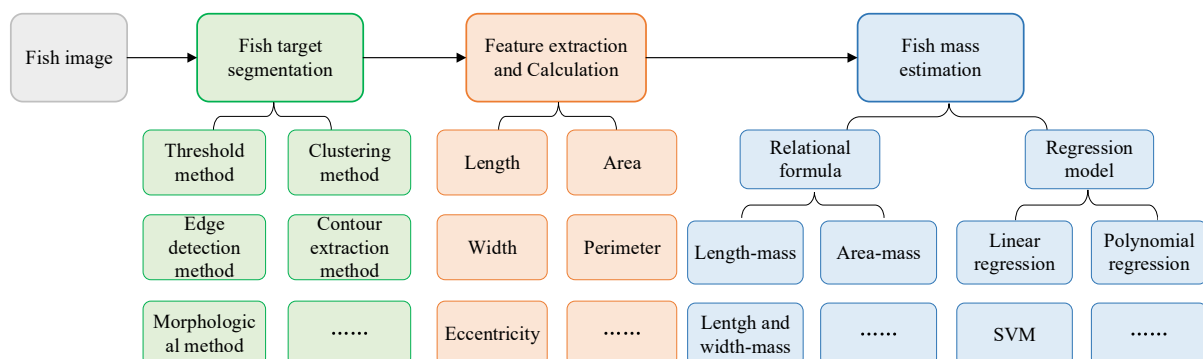


Fig. 11 – Process of fish mass estimation method based on machine vision.

the fish using image processing techniques. On this basis, the actual length of the fish was estimated using the approximate third-order regression curve method. Unlike this method, which obtains only a single-angle image of the fish, the front and side images of the small fish population were obtained by the dual-synchronous orthogonal network camera in the method proposed by Al-Jubouri et al. (2017). Some image processing operations were employed to identify the fish body contour to obtain the length of the fish in the two types of images, and the actual length of the fish was measured using the mapping formula between the image size and the actual size.

Based on the acquisition of fish length, the estimation of fish mass is realised in combination with the length-mass relationship. Martinez-de Dios et al. (2003) also acquired images of fish from different angles using the stereo vision system. In this method, the fish images were preprocessed using noise filtering and illumination compensation technology, the fish features were extracted from different angle images, and the feature points detected in each angle image of the fish were employed as interest points. The fish target was segmented by the method of associating the same fish interest point to obtain the fish length and the fish mass was estimated by the relationship between the length and the mass. The images taken from multiple angles can reflect more comprehensive information about the fish, which helps to estimate the length and mass of the fish more accurately. However, it can be complicated to set up the environment and prepare the imaging equipment in the multi-angle method. Gümüş et al. (2021) solved this problem using a lightbox. In their study, the fish lengths were determined according to the three shapes of the fish placed in the lightbox, namely horizontal placement, placed at a certain angle but the fish body was still straight, and the fish body was curved or wavy. If the fish was placed in the lightbox horizontally and straight, the length of the fish was determined by the leftmost and rightmost pixel values. If the fish is not placed horizontally, but at an angle and still straight, a 'best rectangle' can be fitted to the fish, and the fish length can be measured on this rectangle. If the fish is not 'straight' but is bent or wavy, then the centreline of the fish can be found, and the length of this centreline can be measured. The fish mass was estimated by the relationship between the length and the mass. Instead of installing multiple imaging devices, three shapes of the fish placed in the lightbox reduce the complexity of multi-angle imaging.

Area usually refers to the visible area of the fish that is represented in the image. Currently, some studies estimate the fish mass by constructing the relationship model between the area and the mass. Gümüş and Balaban (2010) and Hufschmied et al. (2011) are two of these studies. The difference is that the former used the out-of-water fish body as the object of the study, while the latter directly collected images and estimated the mass of the fish swimming in the water, which is a more non-destructive and automated study. Furthermore, another interesting point worth discussing can be found: it is known from actual production that fins make up part of the fish's body area, but fins have less influence on the whole fish mass. Would it be more helpful for the estimation accuracy if the mass is estimated based on finless images? On this topic, Kononov et al. (2019) conducted a comparative

study of mass estimation based on whole fish and finless images of Asian perch. The LinkNet-34 segmentation model was trained for these two types of images to segment the fish. On this basis, the area features of each of the two image types were extracted and the mass of Asian perch mass was estimated by constructing a regression model. The experimental results show that the whole fish images were better in estimation. However, this conclusion was only drawn from the study of extracting an area feature to estimate the mass of the Asian perch. The advantages and disadvantages of the performance of whole fish images and finless images need to be further investigated for other fish species and the extraction of other features.

6.1.2. Fish mass estimation method based on multiple factors

The multiple-factor fish mass estimation method explores the relationship between fish mass and several fish features, such as length, area, and perimeter, and establishes a relationship model between fish mass and several features to realise fish mass estimation. Qashlim et al. (2019) calculated the length and width of the fish based on the detected fish contour and realised the estimation of milkfish mass through a calculation formula involving fish mass, length, and width. The accuracy of this method on twenty fish samples was 86.66%, but the calculation formula may not be applicable that resulting in a decrease in accuracy when the fish species were changed. Using machine learning algorithms to build a mass prediction model can effectively avoid the above problems due to its strong fitting ability, which is a research hotspot of fish mass estimation methods based on multiple factors in recent years. Viazzi et al. (2015) conducted a study based on this idea. They obtained an image of Jade perch (*Scortum barcoo*) against a white background, segmented the fish target using the adaptive threshold method, extracted three fish features of the area, length, and width, and predicted the mass using linear regression and polynomial regression. Having said that, it could be concluded that all the above fish mass estimation is implemented based on segmentation from the background to detect the fish. Therefore, the segmentation of fish targets directly affects the effect of fish mass estimation. Although the image background in the Viazzi et al. (2015) study was white and free of other impurities, which greatly reduced the difficulty of segmentation, it was difficult to guarantee the segmentation effect when faced with complex and diverse backgrounds.

To address this problem, Fernandes et al. (2020) employed a deep learning algorithm to segment the target fish, which was able to handle complex and diverse background images. Based on the segmented fish, the features of area, length, height, and eccentricity were extracted, and the mass of the Nile tilapia was estimated by feature fitting. Generally, the frontal image of the fish, that is, the image of the abdomen was used to conduct fish mass studies. From these frontal images, some features closely related to mass such as fish area, body length, and body width are extracted, but information such as fish thickness which is also important for mass estimation is not available. To obtain this information, images from other angles need to be collected. In the study by Saberioon and Císar (2018), fish mass estimation was performed on the dorsal images of the fish. Eight dorsal geometric features were

extracted, and the fish mass was obtained using SVM and decision tree algorithms.

Table 3 concludes the fish mass estimation method based on machine vision.

6.1.3. Summary

Single-factor fish mass estimation methods based on body length and area are simple and easy to implement. However, fish phenotypes are often not uniformly fixed, which not only includes the inherent phenotypic differences of different fish species but also the movement phenotypic differences caused by behavioural changes of the same fish species, making it difficult for a single fixed feature to always accurately describe their morphology and quantify their mass. The multi-factor-based fish mass estimation method can more comprehensively describe the fish morphology by using multiple features, but the extraction of multiple features is currently generally limited to the two-dimensional plane currently, based on which the accuracy of the estimated fish mass needs to be further improved. Table 4 compares the above two types of methods in terms of principles, advantages, and disadvantages.

To further obtain fish mass accurately, it is a research idea to acquire fish image data by a binocular camera and multi-angle cameras, construct a three-dimensional model of fish, extract fish features in a holistic and multi-angle way to describe fish morphology and perform fish mass estimation. In addition, to further improve the practicality of fish mass estimation in actual production, it is necessary to obtain multi-angle images of fish school swimming in the water, build a group diorama model, and directly estimate the overall mass of fish. The estimation study of the overall fish mass not only avoids the difficulty of estimating the mass of an individual fish based on morphological characteristics due to the overlapping of fish but also makes it convenient to obtain the mass by one-time estimation and reduces the cumulative error.

6.2. Fish counting

Counting refers to the estimation of the number of objects in a target area, then the information is obtained, and timely control operations were carried out, which has been widely used in many fields. By studying fish counting, the number of fish in the breeding area can be obtained and knowing the breeding density, which is a key factor to grasp the aquaculture situation, a basis for determining the feeding amount can be provided. The survival rate of the cultured fish is calculated based on the number of fish obtained and the number of fries originally cultured in the breeding area, to assess whether the breeding method is appropriate, whether fish are dying by scavenging each other due to inadequate feeding, whether juveniles are dying because they do not scavenge for food for a long time, or whether weakfish are unable to grow due to the water pollution caused by overfeeding. Furthermore, the mass of all the fish in the nursery is obtained by combining the number of fish with the mass of the fish, and then the feeding amount is initially determined by the feeding rate. Fish counting is therefore the key link in the realisation of intelligent feeding and healthy fish farming.

Fish counting methods based on machine vision have the advantages of high efficiency, low cost, and ease of use, and can effectively avoid the problems of slow speed, error-prone and easy damage to the body surface of manual counting. It has become an important means of research counting. They are based on the collection of fish images by image acquisition equipment, and then fish counting is performed by image analysis. It can be divided into the following methods, as shown in Fig. 12.

6.2.1. Fish counting method based on traditional image processing

Traditional image processing is widely used in fish counting, providing a non-destructive, non-contact method. It mainly includes the image segmentation-based method and the region regression-based method.

The image segmentation-based fish counting method uses threshold, edge detection, and other techniques to segment the target object, processes the segmentation result to determine the target individual, and the number of individuals obtained is the counting result. Therefore, image segmentation is the key technology of this kind of method. Many scholars have conducted fish counting research based on image segmentation methods, as shown in Table 5. According to the different growth stages and sizes of the research objects, it can be divided into a normal fish counting study and a fish egg counting study. In normal fish counting, the size of the target fish is large, so it is easy to distinguish it from noise when determining the target individual. However, due to many behaviours, such as swimming and feeding, fish have different shapes, which can create some obstacles to counting. When counting fish eggs, the objects to be counted are dense, tiny, and individual in shape, making it difficult to distinguish them from noise. Therefore, both pose some challenges for counting. Overlap also poses some difficulties for counting because the image segmentation-based method counts by identifying the number of the fish-connected area. For overlapping images, the problem of less counting may occur when directly identifying it as a single fish since the segmented connected area may contain multiple fish. Therefore, this class of methods is mainly suitable for non-overlapping image counting.

The region regression-based fish counting method directly estimates the total number of targets in the region by building a regression model of regional features and target numbers. It focuses on the total number of fish in an image rather than accurately detecting their specific positions. It is therefore more suitable for counting fish images with significant overlap. Zhu (2009) and Zheng and Zhang (2010) used this type of method to conduct fish counting research, but the overlap in their experimental images was only between two fish. In the method of Fan and Liu (2013), fish images were preprocessed using image processing techniques, seven geometric features of the fish image (e.g., area, perimeter, and convex area) were extracted, and fish counting models were built based on the neural network and the least squares support vector machine separately. Even better, their method was effective for overlaps of up to six fish. For more complex overlapping fish, Zhang, Li, et al. (2020) proposed a method based on image density grading and local regression. The method extracted

Table 3 – Fish mass estimation methods based on machine vision.

Reference	Image processing method	Feature	Mass estimation method	Research target	Results
Martinez-de Dios et al. (2003)	Noise filtering and illumination compensation	Length	Relationship between fish length and mass	Fish in farms	Errors: 4% (nursery tanks), 5% (sea cages)
Balaban et al. (2010)	Image correction	Visible area, length, and width of the best-fitting rectangle	Polynomial regression model	Four different Alaskan salmon and Alaskan pollock	R^2 : 0.953–0.987
Gümüş and Balaban (2010)	—	Area	Linear regression analysis between actual area and mass	Rainbow trout	R^2 : 0.98(linear), 0.99(power), 0.98(polynomial)
Hufschmied et al. (2011) Zhang et al. (2011)	Outline extraction method Grayscale, binarisation, and contour extraction technology	Area Length, width, projected area (head, abdomen, tail), perimeter, and length-width ratio	Linear regression method A polynomial regression model of the projected area of the head, abdomen, and tail	Swimming fish Eighty-six crucian carp	RAE: 5.5% R^2 : 0.9878, MRE: 3.889%, MAE: 6.81g
Wang et al. (2012)	Clustering method	Area	Fitting model of area and mass	<i>Cynoglossus semilaevis</i>	RAE: 5.50%–6.17%
Ault and Luo (2013)	First-order generalised linear model to obtain the fork length	Perimeter and fork length	Allometric growth model, wood geometry formula, and generalised multivariate linear statistical model	A public dataset of Atlantic Tarpon	R^2 : 0.8876–0.9997
Costa et al. (2013)	Image analysis and contour morphometry	Area, major and minor axis length, and perimeter and centroids size	Least squares support vector regression	Perch	R^2 : 0.9772
de Verdal et al. (2014)	Converting the image to black and white, adjusting the brightness and contrast, adjusting the threshold, filling holes, and analyzing pixel particles.	Area, perimeter, length, height, and volume of the side image	Two partial least squares discriminant analysis models	European sea bass larvae	R^2 : 0.98
Viazzi et al. (2015)	Target detection, adaptive threshold segmentation, and tail fin removing	Area, length, and width	Linear regression and polynomial regression methods	Jade perch <i>Scortum barcoo</i> in aquaculture farms	R^2 : 0.99, MRE: $6 \pm 3\%$
Gerami et al. (2016)	Colour model extraction, median filter, grayscale, and noise reduction	Length, height, area, perimeter, equivalent diameter, major and minor length	Linear regression and multiple regression	Seventy-five <i>Oncorhynchus mykiss</i>	R^2 : 0.784–0.998
Saberioon and Císař (2018)	Five replicate calibration, background image subtraction, thresholding, and the morphological operation method to segment, and image alignment	Area, length, maximum width, and five equidistant fish widths	SVM and random forest (RF) prediction models	Perch	RF: R^2 : 0.868, RMSE: 0.1300; SVM: R^2 : 0.872, RMSE: 0.1304
Sanchez-Torres et al. (2018)	Homomorphic filtering, adaptive histogram equalisation with limited contrast, two-dimensional saliency detection, and morphological operators	Length	Polynomial regression	One hundred and fifty <i>Oreochromis niloticus</i>	Error: 11.24% (grade-3 polynomial) 11.88%: (grade-4 polynomial)
Konovalov et al. (2019)	LinkNet-34 segmentation model	Area	Regression model	Asian perch	MAPE: 4.36% (whole-fish), 4.28% (no-fins)

Qashlim et al. (2019)	Grayscale processing, canny method, and threshold feature method	Length and width	Calculation formula that involves fish mass, length, and width	Milkfish	Accuracy: 86.66%
Fernandes et al. (2020)	Deep learning algorithm to segment the target fish	Area, length, height, and eccentricity	Linear fitting	Nile tilapia	R^2 : 0.96
Zhang, Wang, and Duan (2020)	Image scaling, Grabcut method for segmentation, flood filling, median filtering, and adaptive histogram processing for enhancement, and Otsu for binarisation	Area, perimeter, length, width, equivalent diameter, Heywood circularity factor, solidity, complexity, and arc.	BPNN	Crucian carp	MAE:0.0104, RMSE:0.0137, R^2 : 0.9021
Gümüs et al. (2021)	–	Length	Relationship between fish length and mass	European catfish and African catfish	R^2 : 0.927–0.972
Abinaya et al. (2022)	YOLOv4 for fish segmentation, the nearest neighbor algorithm to associate the fish segments	Fish head, fish body, and fish tail	Calibration curve	Tilapia	MAE: 4.41, MARE: 5.85, Accuracy: 94.15, R^2 : 0.9861
Hao et al. (2022)	Active contour model for segmentation and a method of Cartesian coordinate system and image processing for fish tail fin removing	Fish body area, length, and height from fish with tail fin and body area, length and height from fish without tail fin	Multiple linear regression models based on the Partial Least Square	Crucian carp (<i>Carassius auratus gibelio</i>)	R^2 : 0.991, RMSE: 7.10, MAE: 5.36, MaxRE: 8.46%
Yu et al. (2022c)	The proposed method called CAM-Decoupled-SOLO for fish segmentation	Perimeter	Linear, power curve, and square root models	<i>Oplegnathus punctatus</i>	R^2 :0.973,0.966,0.970; RMSE: 24.51,27.84,25.15; MAE:19.94,20.46, 19.46; MAPE:7.66,6.64, 6.54
Note: R^2 : coefficient of determination, MAPE: mean absolute percentage error, MAE: mean absolute error, RMSE: root mean square error, RAE: relative average error, MRE: mean relative error, MaxRE: maximum relative error, MARE: mean absolute relative error.					

Table 4 – Comparison of fish mass estimation methods.

Method	Category	Principle	Advantages	Disadvantages
Single-factor methods	Formula calculation	Construct calculation formula of single feature and mass	Simple principle, no need for a large amount of data for model training, easy to implement	Only be applied to specific species of fish, with weak universality
	Regression model	Construct a regression model of single feature and mass using a machine learning algorithm	Enable to realise mass estimation of different fish species	A single feature cannot fully describe fish of different shapes
multiple-factor methods	Formula calculation	Construct calculation formula of multiple features and mass	No need for a large amount of data for model training, easy to implement; multiple features can describe the shape of the fish more comprehensively	The more features, the more complex the calculation formula, and the more difficult it is to fit
	Regression model	Construct regression model of multiple features and mass using machine learning algorithm	Can accurately estimate the mass of different species of fish	Correlation and coupling may exist between multiple features, and the estimation accuracy needs to be improved

features from these segmented fish-connected areas. All the fish-connected area images were graded into a density, such as low density, medium density, and high density, to build three datasets, based on which three models were trained to perform the fish counting on the connected area images in each density. Finally, the number of fish images was obtained by merging the counting results of the connected areas belonging to the same image. This method provides a more accurate and stable counting for highly overlapping fish images.

6.2.2. Fish counting method based on deep learning

Deep learning is powerful at representing data. It can automatically combine low-level features to form more abstract high-level representational attribute categories or features, which has greater adaptability. Using deep learning algorithms for fish counting can avoid image segmentation with complex backgrounds and manual feature extraction, making the counting model more universal (Zhang, Zhou, et al., 2022). In deep learning fish counting, it can be divided into the image segmentation-based method, the density estimation-based method, and the target detection-based method.

The image segmentation-based method uses the deep learning algorithm to automatically segment the target object from the background and realises the counting by feature thresholding setting or feature regression. Tseng and Kuo (2020) used a mask region-based CNN method to detect and segment harvested fish from the background at the pixel level in electronic monitoring system video frames. The number of fish was determined using time and distance thresholding methods. The counting precision was low at 77.31%, which can be influenced by the counting results obtained by the thresholding method. To address this issue, Nguyen et al. (2020) conducted a study in which counting was performed using the constructed regression model. The ResNet-50 and feature pyramid were used to build a segmentation model, model training and validation were performed using the pre-trained weights, and the proposed two-stage prediction method was employed to realise shrimp larvae counting. This method had a good performance on the low-overlapping and medium-overlapping shrimp images. The density estimation-based method uses the deep learning algorithm to map the input image into the corresponding density map and realises the counting based on the summation of mathematical integrals (Yu et al., 2022b; Zhang et al., 2020). The performance of this class of methods depends to some extent on the quality of the density map. The target detection-based method builds a detection model, trains it based on the dataset, and detects the objects in the image using the optimal model obtained from training. The number of detected objects is then the counting result.

Xu et al. (2018) trained a Faster R-CNN underwater target detector to obtain target boundaries and categories and trained a kernelized correlation filter tracker through the detector to achieve the identification and counting of various seafood such as sea cucumbers, sea urchins, scallops, etc. Fast R-CNN is a two-stage target detection algorithm, in the detection process of which, the completion of sample classification and identification is in order. Therefore, this type of method has a slower speed. Another type of single-stage

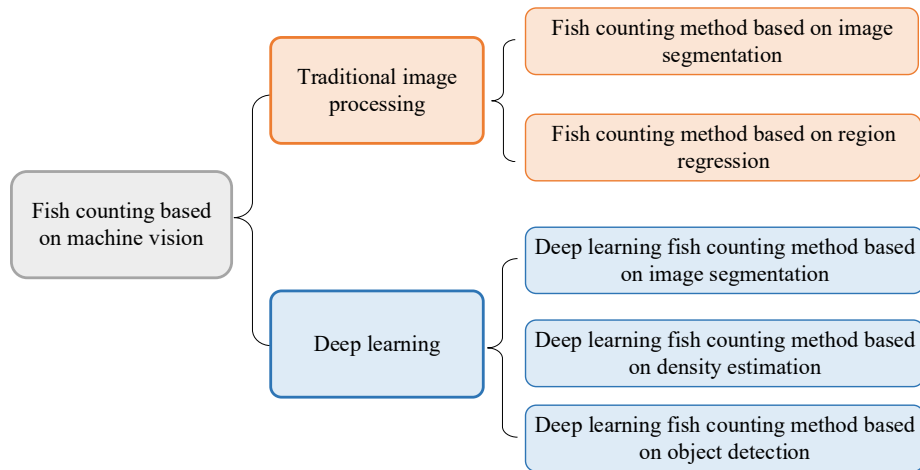


Fig. 12 – Classification of fish school counting methods based on machine vision.

target detection algorithm does not need to generate candidate images. It converts the target positioning problem into a regression problem and completes target positioning and identification simultaneously with low time complexity. Thus, in the methods of [Kandimalla et al. \(2022\)](#), [Cai et al. \(2020\)](#), and [Zhang, Huang, et al. \(2021\)](#), the single-stage target detection algorithms YOLOv3 and YOLOv4 were used for fish counting, which improved the speed. A comparison of the fish counting methods based on deep learning is shown in [Table 6](#).

6.2.3. Summary

The fish counting method based on traditional image processing does not need a large number of labelled datasets to train the model but counts based on the segmentation of the target fish school, inheriting the drawbacks of traditional image segmentation algorithms. Its counting performance depends on the effect of segmentation. The fish counting method based on deep learning can process multi-background images with strong model robustness, but the counting efficiency needs to be improved. [Table 7](#) shows the comparison of the above fish counting methods.

In real aquaculture, fish have a variety of behaviours such as feeding, swimming, and abnormalities, with different expressions. For example, fish that are feeding appear to congregate at the feeding site, resulting in a large number of fish in that area of the image, whereas fish that are not feeding are scattered in several areas where the number of fish is low. As a result, the density of fish in different areas of the same image is uneven. There are also different degrees and forms of overlap between fish in different density areas. Therefore, it is helpful to further realise fish counting accurately and efficiently by differentiating different density levels and constructing lightweight counting models separately. In addition, it is worth exploring the idea of combining multi-frame counting of videos, as the overlapping phenomenon makes the accuracy of counting based on a single static image not high.

6.3. Fish feeding behaviour analysis

In the process of feeding, fish perceive the signal of feed feeding, and produce a range of feeding behaviours based on their appetite. In other words, feeding behaviour reflects the appetite of the fish. Studies have shown that fish with different appetites exhibit different feeding behaviours ([Eriksen et al., 2011](#)). For example, the behaviour of not responding to feed corresponds to non-feeding; the behaviour of consuming only the feed in front of them corresponds to weak feeding; the behaviour of swimming to the feeder to consume the feed and then returning to the starting point corresponds to feeding; the behaviour of moving irregularly at the feeder and consuming all the feed corresponds to strong feeding, etc. Therefore, it is important to monitor and analyse the feeding behaviour of fish to capture the intensity of feeding activity to adjust the feeding amount promptly, reduce the breeding cost and improve the breeding efficiency ([An et al., 2020](#); [Zhao et al., 2021](#)).

The feeding behaviour analysis method based on machine vision extracts the features that can reflect the feeding behaviour from the captured videos or images and constructs an identification model or quantitative indicators to realise feeding status recognition, which has the advantages of objectivity and non-destructive. According to the way of feature extraction, it can be divided into the traditional image processing-based method and the deep learning-based method, as shown in [Fig. 13](#).

6.3.1. Fish feeding behaviour analysis method based on traditional image processing

The feeding behaviour analysis method based on traditional image processing segments the target, manually extracts the features and obtains the feeding status of the target by machine learning models or quantitative indicators. It is a common feeding behaviour analysis method to construct an identification model that combines the manually extracted texture, shape, and other features with machine learning

Table 5 – Fish counting method based on image segmentation.

Reference	Image segmentation algorithm	Target counting method	Research target	Results
Toh et al. (2009)	Background difference method	Determine the total number of fish by the area value	Fish school	Percentage Error: 0% (5 or 10 fish), 0.19% (15 fish), 0.07% (50 fish)
Labuguen, Volante, and Causo (2012)	Adaptive threshold method, and edge detection algorithm	Determine the total number of fish by the area value	Fish school	Accuracy: 68.10%–99.50%
Al-Kofahi et al. (2013)	Two-dimensional wavelet algorithm	Determine the individual using the local maximum detection algorithm	Fish cell	Accuracy: 81%
Del Río et al. (2013)	Roberts edge detection algorithm	The number of connected areas segmented	Fish school	–
Fabic et al. (2013)	Canny edge detection algorithm	The number of connected areas segmented	Fish school	–
Fier et al. (2014)	Background difference method	Target tracking and set height, width, area, and hue thresholds for individual identification	Fish school	Sensitivity: 77.9%, precision: 83.8%
Westling et al. (2014)	Background difference method	A machine learning algorithm for individual recognition	Fish school	mAP:0.73
Duan, Stien, et al. (2015)	Otsu algorithm, and improved watershed algorithm	The number of connected areas segmented	Egg cells of American haddock and Atlantic cod	Average counting error: 2%
Al-Saaidah et al. (2018)	Canny edge detection algorithm	Set colour and shape feature thresholds to determine the individual	Zebrafish egg cell	Average detection error: 17.5%
Hernández-Ontiveros et al. (2018)	Threshold method	Set area and perimeter thresholds to identify the individual	Ornamental fish school	Average accuracy: 96.64%
Rahmadiansah et al. (2018)	Background difference method	Find the target centroid to determine the individual	Fish school	Average error: 1.0%
Aliyu et al. (2020)	Edge detection method	The artificial neural network to recognise the catfish	Catfish	Accuracy: 100% (on two samples)
Awalludin et al. (2020)	Image enhancement, edge detection, and thresholding methods	Setting the feature threshold to determine the centroid point, and its number is the result of fish counting	Nile Tilapia	Percentage error: 0.31% (small sample), 1.99% (medium sample), 1.36% (large sample)
Zhang, Pang, et al. (2021)	Otsu algorithm	Connected area labeling algorithm and thinning algorithm to identify fish fry	Fry	Correct rate: 85.7%
Zhou et al. (2022)	Adaptive threshold segmentation algorithm	Segment and detect the fingerlings	Five different fingerlings	Average counting accuracy rate: >98.78%, standard deviation: <0.95%

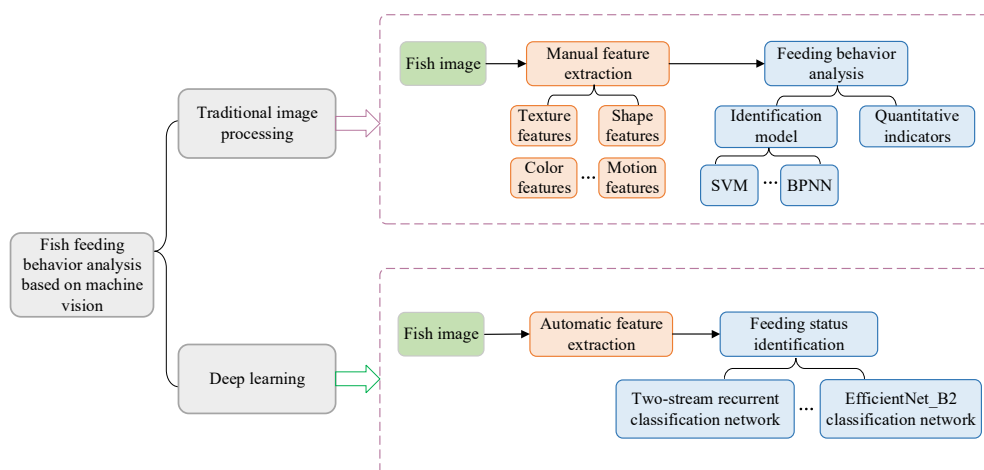
Table 6 – Fish counting method based on deep learning.

Reference	Model	Target counting method	Research target	Result
Xu et al. (2018)	Faster R-CNN	Kernelized correlation filter tracker	Sea cucumbers, sea urchins, scallops, etc.	mAP: 79.6%
Lainez and Gonzales (2019 2019)	Convolutional Neural Network	Detection results	Four different size images of Tilapia fry	Average accuracy: 99.17%
Cai et al. (2020)	MobileNetv1 and YOLOv3	Detection results	<i>Takifugu rubripes</i>	AP: less than 80%
Nguyen et al. (2020)	ResNet-50 and feature pyramid	Two-stage prediction method	Shrimp larvae	Accuracy: 99.7% (low-overlapping), 92.2%–95.4% (medium-overlapping), 72.9% (high-overlapping)
Tseng and Kuo (2020)	Mask region-based CNN	Time and distance thresholding methods	Harvested fish	Recall: 93.84%, Precision: 77.31%.
Zhang et al. (2020)	Hybrid neural network	Mathematic integration of density map	Fish populations in far offshore Atlantic salmon mariculture	Accuracy: 93.43% (Fewer) 94.21% (Few) 95.77% (Medium) 97.02% (Many) 97.55% (Large)
Zhang, Huang, et al. (2021)	YOLOv3	Detection results	Multi-scale fish with complex background	Precision: 96.26%, Recall: 90.65%, F1 value: 93.37%, mAP: 90.20%
Kandimalla et al. (2022)	YOLOv4	Detection results	Fish of Chinook, Jack Chinook, and Sockeye species	–
Yu et al. (2022a)	A network containing an attention mechanism module and a dense residual module	Fish density map	Goldfish, carp, hairtail, ornamental fish, and others	MAE: 7.96 (low density), 24.78 (medium density), 42.53 (high density)
Yu et al. (2022b)	Deep learning network model based on multi-modules and attention mechanisms	Mathematic integration of density map	Cultured fish in a factory	Accuracy: 97.12%, Deviation: 3.67
Zhao et al. (2022)	An LFCNet model consisting of the encoder, decoder, and generation head	Density map	Carp	MAE: 4.10, RMSE: 6.13
Costa et al. (2023)	Convolutional Neural Network	Detection results	Tilapia larvae	Precision: 0.98 Recall: 0.92 Fmeasure: 0.95 MAE: 1.43
Li et al. (2023)	The multi-column neural network of MCNN and the attention mechanism module of Squeeze-and-Excitation Networks	Fry density maps	Crucian carp fries	MAE: 3.33, MSE: 4.58

Note: RMSE: root mean square error, MAE: mean absolute error, MSE: mean squared error.

Table 7 – Comparison of fish counting methods based on machine vision.

Method	Principle	Advantages	Disadvantages
Image segmentation method	Targets segmented with traditional image processing technologies and counting realised by feature threshold	Simple principle, no data training	Unable to handle complex background and overlapping fish counting
Region regression method	Targets segmented with traditional image processing technologies and counting realised by a regression model	No need to identify each target object	Unable to handle fish counting images with different backgrounds
Image segmentation deep learning method	Targets segmented with deep learning and counting realised through feature threshold	Able to handle fish counting images with complex backgrounds	Require a large amount of data to train the segmentation model and with low time efficiency
Density estimation deep learning method	Image mapped to a density map and counting realised based on the mathematical integral summation	Able to handle dense fish counting	Performance depends on the quality of the density map, prone to cumulative errors
Target detection deep learning methods	Locate and identify targets using a deep learning algorithm	High detection accuracy and counting accuracy	Require a large amount of data to train the segmentation model and with low time efficiency

**Fig. 13 – Classification of feeding behaviour analysis methods based on machine vision.**

methods. It is a supervised method that performs model training based on the labelled data and identifies feeding status by the obtained optimal model. Another unsupervised feeding behaviour analysis method constructs a quantitative index of feeding behaviour from manually extracted features and judges the intensity of feeding activity based on this index. Both of them have some research, as shown in Table 8.

Due to the problems of poor illumination and turbid water in the actual aquaculture environment, the quality of the visible light images obtained can be poor, affecting the effect of identifying fish behaviour. Near-infrared machine vision technology is not affected by the intensity of visible light and can achieve better imaging results in relatively dark environments. Therefore, Zhou et al. (2018) and Zhou, Zhang, et al. (2017) analysed the feeding behaviour based on the obtained near-infrared images of fish schools. On the one hand, the centroid of the fish was calculated using the order moment,

which was used as the vertex for triangulation, and then the fish school index was calculated as a quantitative index of feeding behaviour. The least-square polynomial method was employed to fit the index value, thus quantifying the intensity of fish feeding activity. On the other hand, since the texture features can reflect the characteristics of spatial information such as fish movement direction and variation range, the texture features of the images were extracted by the grayscale co-occurrence matrix to represent the predation degree of the fish school and quantify the fish feeding activity. Both methods were studied separately, while a comparative study of their performance is interesting and may be carried out in the future.

In addition, fish behaviours such as hunger, aggregation, and aggression may also reflect their feeding desires to some extent (Chang, 2005; Taha et al., 2018; Zhao et al., 2016). Therefore, it is also important to identify these behaviours to

Table 8 – Feeding behaviour analysis method based on traditional image processing.

Reference	feature	Main algorithm	Function	Research target	Results
Liu et al. (2014)	The sum of all pixel intensities of consecutive frame differences	Interpolation	Acquire fish feeding activity index	Atlantic salmon	–
Sadoul et al. (2014)	Change the degree of the fish school area and spread the degree of fish school	–	Obtain fish feeding activity, and conclude that the fish speed and direction can also quantify fish feeding intensity	Rainbow trout	–
Wei et al. (2014)	Fish's area, and area ratio	–	Give the formula for calculating the amount of bait	Fish school	–
Hu et al. (2015)	The area ratio of the feature area	Draw the curve of the characteristic parameter value over time	Study the changing characteristics of fish feeding behaviour	Fish school	–
Ye et al. (2016)	Fish speed, and fish turning angle	Combined entropy method	Analyse the feeding intensity of tilapia	Tilapia	–
Zhao, Zhu, et al. (2016)	Change the range of the reflective area	Optical flow method, statistical method, information entropy	Assess the intensity of feeding activities	Tilapia	–
Chen et al. (2017)	Inverse moment difference, correlation, energy, and contrast	–	Analyse the change in the feeding behaviour of fish based on the change rule of characteristics	Fish school	R ² : 0.8942
Chen (2017) Zhao et al. (2017)	Texture feature Fish concentration and dispersion, interaction force, and variation range of water flow field	SVM Covariance, improved social force model, improved kinetic energy model	Identify fish status Get the activity intensity of fish feeding behaviour	Fish school Tilapia	Accuracy: 96.5% Non-match rate: (2.19 ± 0.81)%
Guo et al. (2018)	Shape feature and texture feature	BPNN	Identify fish feeding status	Mirror carp	Accuracy: 98.0%
Chen and Chen (2019)	Texture feature, motion direction feature, and disorder feature	SVM	Identify fish feeding behaviour	Fish school	Accuracy: 97.0%
Zhang et al. (2019)	Colour feature, shape feature, and texture feature	BPNN	Identify fish feeding behaviour	Mirror carp	Precision: 97.1%
Yu et al. (2021)	Feature points and velocities of sub-images	Model combining normal behavioural characteristic matrix and special behaviour detection	Identify fish school feeding and other special behaviours.	Fish school	Accuracy: 96.02%

Note: R²: coefficient of determination.

Table 9 – Feeding behaviour analysis method based on deep learning.

Reference	Model	Feeding status	Research target	Results
Måløy et al. (2019)	Dual-Stream Recurrent Network	Feeding and non-feeding	Salmon	Accuracy: 80%
Zhou et al. (2019)	Convolutional neural network	Feeding intensity of none, weak, medium, and strong	Tilapia	Accuracy: 90%
Saminiano (2020)	Convolutional neural network	Feeding and non-feeding	Tilapia	Precision: 94.87%, Recall: 94.87%, Specificity 97.93%, Accuracy: 96.40%
Ubina et al. (2021)	Optical flow neural network and 3D convolutional neural network	Feeding intensity of none, weak, medium, and strong	Fish	Accuracy: 95%
Yang, Yu, et al. (2021)	Dual attention network and EfficientNet-B2	Before, during, and after feeding	<i>Oplegnathus punctatus</i>	Accuracy: 89.02%
Feng et al. (2022)	ResNet and GloRe	None, weak, medium, strong	Trout	Accuracy: 92.68%
Kong et al. (2022)	VGG16 network	None, weak, medium, strong	<i>Oplegnathus punctatus</i>	Accuracy: 98%
Xu et al. (2022)	Long-term recurrent convolutional networks	Feeding and non-feeding	Grass carp and crucian carp	Accuracy: 97%
Zhang, Wang, et al. (2022)	MobileNetV2 and SENet	Feeding and non-feeding	<i>Plectropomus leopardus</i>	Accuracy: 97.76%
Zheng et al. (2023)	Spatiotemporal attention network	Feeding and non-feeding	Fish	Accuracy: 97.97%

guide feeding. Taha et al. (2018) established the relationship between behaviour and starvation status. The image processing technology was used to analyse the behavioural data of Asian sea bass, and the K Nearest Neighbors (KNN) algorithm was used to classify fish school images, which realised the recognition of fish school starvation behaviour. Chang (2005) observed the aggregation behaviour of eels, converted it into digital signals, and then the feeder connected to the control system controlled the feed feeding according to the analysis results of the aggregation behaviour of eels.

6.3.2. Fish feeding behaviour analysis method based on deep learning

The deep learning-based feeding behaviour analysis method does not require manual extraction of fish image features. In this type of method, a predefined label is used as the result of the fish feeding status analysis, i.e., if there are two predefined feeding statuses, the identification result can only be one of them. In the study by Saminiano (2020), the two predefined labels were feeding and non-feeding. A convolutional neural network algorithm was used to analyse the feeding status of individual fish images with an accuracy of 96.4%. To further refine the intensity of fish feeding activity, Ubina et al. (2021) proposed a two-stage approach to achieve the identification of four different levels of feeding intensity: none, weak, medium, and strong. In the first stage, an optical flow neural network was applied to generate optical flow frames, and the generated optical frames were fed into the 3D convolutional neural network to achieve the identification of four different levels of feeding intensity in the next stage. Instead of fish images, optical flow frames obtained from videos were used as input images for the feeding behaviour recognition network, and the accuracy reached 95%. However, the construction and training of the two deep network models

consumed a lot of time, resulting in a decrease in recognition efficiency. A comparison of the feeding behaviour analysis method based on deep learning is shown in Table 9.

6.3.3. Summary

The feeding behaviour analysis method based on traditional image processing requires manual feature extraction, and the generalisation and adaptability of the model are not strong. Thus, it is only suitable for image data with a single background collected in the laboratory environment and its application scope is limited. In this regard, the deep learning-based feeding behaviour analysis method performs better, but the balance between the accuracy and efficiency of deep learning models still needs to be explored. Currently, most of the fish feeding behaviour analysis studies focus on the identification of fish feeding behaviour states, such as feeding, strong feeding, weak feeding, non-feeding, etc. It lacks a more granular breakdown of appetite during the fish feeding and is unable to fully quantify the overall feeding appetite of the entire fish feeding process. In addition, since fish live in a three-dimensional space of water, changes in feeding behaviour also lead to changes in their spatial distribution and structure. Therefore, it is an important idea to quantify the dynamic appetite of fish in the whole feeding process in combination with the spatial distribution and structural characteristics, which can help to realise the analysis of fish feeding behaviour more accurately and in detail.

6.4. Aquaculture water feed detection

Compared to the direct analysis of fish feeding behaviour, the feed in aquaculture water indirectly reflects the feeding situation of the fish, which is one of the most important methods to assess the intensity of fish feeding activity. The amount of

Table 10 – Aquaculture water feed detection method based on machine vision.

Reference	Image processing	Detection method	Function	Result
Atoum et al. (2015)	–	Correlation filter for feed detection	Continuously monitor fish eating activity, detect excess feed, and automatically control the feeding process	Detection rate: 90.80%
Skøien et al. (2015)	Image blurred, and background and pellet separation	Blobs identified and located	Detect and quantify feed particle	Error: 1.3%
Wang (2016)	Image denoising, gray enhancement, morphological processing, and image segmentation	Least squares support vector machine model to identify the bait	Calculate the fish feeding activity index and adjust the feed amount	Relative error: <5%
Li et al. (2017)	Histogram fitting	Adaptive threshold method to segment the feed foreground	Detect uneaten fish food underwater	True Positive Rate: 80% ~95.90%, False Positive Rate: <2.7%
Cao and Xu (2018)	Image dehazing, image restoration, and image enhancement	Threshold method to identify the contour and estimate the number of feed pellets based on individual pellet area	Identity and count residual feeds	Relative error: around 10%
Hu et al. (2021)	Image and data enhancement	Detection model based on the improved YOLOv4 network	Real-time detection of uneaten feed pellets in underwater images	AP ₅₀ : 92.61%
Hou et al. (2022)	Image resized	The improved multi-column convolutional neural network to detect	Detect feed particles	MAE: 2.32, MSE: 3.00
Wang et al. (2022)	–	Multi-task convolution neural Network to detect uneaten feed pellets	Analyse the feeding activity of cultured fish, monitor the number of uneaten feed pellets, and dynamically adjust feeding	MAE: 4.80, MSE: 6.75

Note: MAE: mean absolute error, MSE: mean square error.

residual feed reflects the intensity of fish feeding activity. For example, if the amount of residual bait increases in the same period, it indicates that the intensity of fish feeding activity is weakening. In addition, the residual bait in the aquaculture water not only increases the cost of aquaculture, but also may pollute the aquaculture water and cause fish diseases (Da Silva et al., 2016; López-Olmeda et al., 2012; Skøien et al., 2016; Zhang et al., 2012). Therefore, it is necessary to conduct research on feed detection to obtain the amount of residual bait in aquaculture water, capture the feeding intensity of fish, and dynamically adjust the feeding amount to achieve intelligent feeding in aquaculture.

6.4.1. Feed detection method

In the past, some researchers have used acoustic technology for feed detection research (Juell, 1991; Llorens et al. 2017). However, the high cost of acoustic technology and its susceptibility to noise interference limited its application in practical production. Machine vision technology has become the focus of feed detection research due to its advantages of being no-destructive, low cost, and convenient to use. By capturing images during fish feeding, detecting the amount of feed in the image, and analysing the intensity of fish feeding activity, it is possible to decide whether to continue feeding or not, thus realising the intelligent feeding of fish. Cao and Xu

(2018) used image processing technology to enhance the underwater feeding image and obtained the feeding contour. The contour was identified whether it was a single pellet or a feed pile based on the threshold method, and the number of feed pellets was estimated based on the obtained area of the single pellet. In this method, each feed particle has to be segmented for counting. However, the feed particles are small and dense and may be occluded by fish, which can cause some difficulties for segmentation-based counting. Deep learning feed detection doesn't need to segment each feed and avoids this problem very well (Hou et al., 2022; Hu et al., 2021). However, training deep learning detection models requires a large amount of labelled data, and labelling a single grain of feed will require a lot of effort. Table 10 summarises the aquaculture water feed detection method based on machine vision.

6.4.2. Summary

The boundary contour of the feed individual is somewhat blurred, and the boundary between feed and background is not obvious due to the characteristics of small feed individuals, dense groups, fish interference, swimming movement, etc. As a result, it is difficult to detect, segment and manually label feed individuals. In addition, the detection of residual feed in the aquaculture water is a task with high real-time requirements, as the time delay is not conducive to

subsequent feeding feedback control. Therefore, it is worth exploring the idea of identifying and counting the residual feed in the aquaculture water as a whole, which not only avoids the problem of fuzzy identification of individual feed boundaries but also helps to improve the identification efficiency. Furthermore, to improve the practicality of using feed detection to guide feeding, it is necessary to investigate the relationship between the number of residual feeds and the feeding intensity of the fish.

6.5. Intelligent feeding decision

Feeding is an important factor in aquaculture that determines the efficiency and cost of aquaculture, and is important for promoting the healthy growth of fish and reducing the cost of aquaculture. Currently, feeding in aquaculture is mainly based on mechanical timing and quantity, relying on a fixed amount of feed preset by the breeder, which cannot be automatically adjusted according to the feeding needs of the fish, resulting in the problems of underfeeding or overfeeding, leading to reduced growth efficiency of fish, feed waste, and water pollution. As fish feed intake is affected by various factors such as water temperature, fish mass, and culture density, the traditional feeding method based on farmers' experience cannot be inherited to accurately meet the fish feed demand at all times. Therefore, it is urgent to research intelligent feeding methods that are sensitive to environmental conditions or growth status.

6.5.1. Feeding decision method

Currently, there are three main types of intelligent feeding methods in aquaculture: mathematical model-based method, acoustic technology-based method, and machine vision-based method (Zhou, Xu, et al., 2017).

The mathematical model method refers to using data models to monitor and predict fish feed consumption, and then formulate feeding strategies. This type of method predicts fish feed intake according to specific factors, mainly including the methods of empirical growth model (Li et al., 2021) and bioenergy model (Chen et al., 2020; Kitchell et al., 1977) methods. Empirical growth models and bioenergetic models tend to achieve good accuracy in a stabilised aquaculture environment, but the performance of the models will

decrease when factors such as farming mode, feed type, etc. change. The acoustic technology-based method uses acoustic sensors to measure feeding amount or fish behaviour to decide when to stop feeding. According to different methods of acoustic data acquisition, it can be divided into biotelemetry method (Sarker et al., 2013), passive acoustic method (Alavandan & Baboo, 2012) and active acoustic method (Tarling et al., 2021), which have problems such as fish body damage, high cost of data collection and vulnerability to environmental noise. A schematic diagram of the feeding control method based on passive acoustics is shown in Fig. 14.

The machine vision-based method refers to using machine vision technology to obtain the physiological information of the fish and the feeding demand information. Then, the feed feeding is controlled based on the analysis results. Zhou et al. (2018) obtained the feeding demand information based on the analysis of fish feeding behaviour and carried out the automatic control of feed feeding. Based on the collected fish feeding behaviour images, the fish school predation and fish school index were extracted by image processing technology as quantitative indicators of fish feeding behaviour intensity. The feeding strategies were then formulated according to the quantitative results to control the feeding. On the other hand, Hu et al. (2022) obtained the feeding demand information based on aquaculture water feed analysis. Unlike most of the previous works, which focused on the image-based analysis of the feed residue on the pond bottom to determine whether to continue or to stop feeding, they used deep learning to detect the size of the waves caused by fish feeding to adjust the feeding strategy. Different from the above single model-based feeding guidance, Wang et al. (2022) integrated the results of feeding behaviour analysis and aquaculture water feed analysis to propose an intelligent feeding method for fish cultured in recirculating aquaculture systems. The features of the input image were extracted by the shared front segment module, the feeding activity of cultured fish was analysed, and the number of uneaten feed pellets was monitored based on the proposed multi-task network. Then the feeding amount was adjusted dynamically. It is more reasonable to make feeding control when the fish feeding demand is analysed from multiple perspectives.

6.5.2. Summary

At present, intelligent fish feeding using machine vision is still at an early stage of research, mainly analysing the fish feeding intensity based on images or videos. On the one hand, the fish feeding behaviour analysis technology is being studied based on the characteristic that fish feeding behaviour varies with the intensity of feeding activity. On the other hand, the aquaculture water feed detection technology is being studied based on the fact that the amount of feed also reflects the intensity of feeding activity. It is hoped that the feeding decision can be made by detecting the intensity of the feeding activity of the fish. However, the current research remains at the stage of feeding activity intensity analysis and fails to further combine the obtained feeding activity intensity with the specific feeding amount, feeding frequency, and feeding time to make a complete feeding decision that can truly meet the actual production application. In addition, combining the

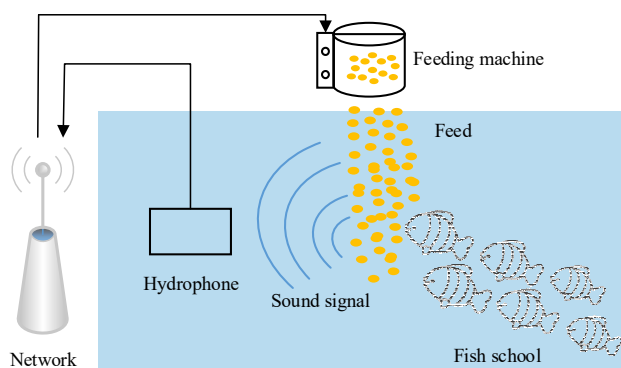


Fig. 14 – Schematic diagram of feeding method based on passive acoustics.

physiological information of fish to make feeding decisions will help to further promote accurate feeding.

7. Challenges and potential solutions

The goal of intelligent fish feeding based on machine vision is to use machine vision technology to analyse fish feeding demand, automatically adjust the feeding amount, provide optimal growth conditions for fish, reduce breeding costs, and improve breeding efficiency. Intelligent feeding aims to provide a fast, non-destructive, and automatic way to feed fish without human intervention. More importantly, it should provide some support for actual aquaculture, freeing aquaculture from reliance on artificial experience and creating a systematic, standardised, and intelligent farming production line. However, many challenges need to be overcome before these goals can be achieved, as shown in Fig. 15. The challenges and potential solutions of intelligent fish feeding based on machine vision are discussed from the data, model, and application perspectives in this section.

7.1. Data perspective

7.1.1. Difficult to obtain multi-source data

Various data such as fish behaviour data, aquaculture environment data, meteorological environment data, and fish physiology data are involved in aquaculture. To collect these data, it is necessary to install cameras, use several sensors, cooperate with the meteorological office, etc. As a result, a lot of manpower and funds are in need. Despite this, some

environmental data that have a greater impact on fish growth, such as ammonia nitrogen, are not yet fully automated. At the same time, it was found that changes in fish physiology, nutritional requirements, environmental conditions, and other factors are directly reflected in their behaviour. Therefore, through changes in fish behaviour, the impact of changes in hard-to-obtain data on fish nutritional requirements and guide feeding can be indirectly estimated, thus avoiding the problem of difficult access to multi-source data.

7.1.2. Uneven data quality

The real aquaculture environment is uncontrolled. Algae, impurities, and unknown objects block the lens, so the image and video data obtained cannot capture the full view of the fish. In addition, the uneven light intensity, turbid water, low visibility, and severe environmental noise make the quality of the acquired image and video data uneven, leading to difficulties in data processing and affecting automated analysis such as fish counting, fish behaviour analysis, and feed detection. Therefore, a series of operations such as image de-drying, image enhancement, and colour space information analysis should be performed to improve image quality before automated analysis.

7.1.3. Time-consuming and laborious manual labelling

Deep learning technology has advantages in processing fish images with complex and varied backgrounds and different target sizes. However, it is a data-based model that requires a large amount of manually labelled data to train, which takes a lot of time and energy. For example, in counting research, every object in the image needs to be labelled. If the counting

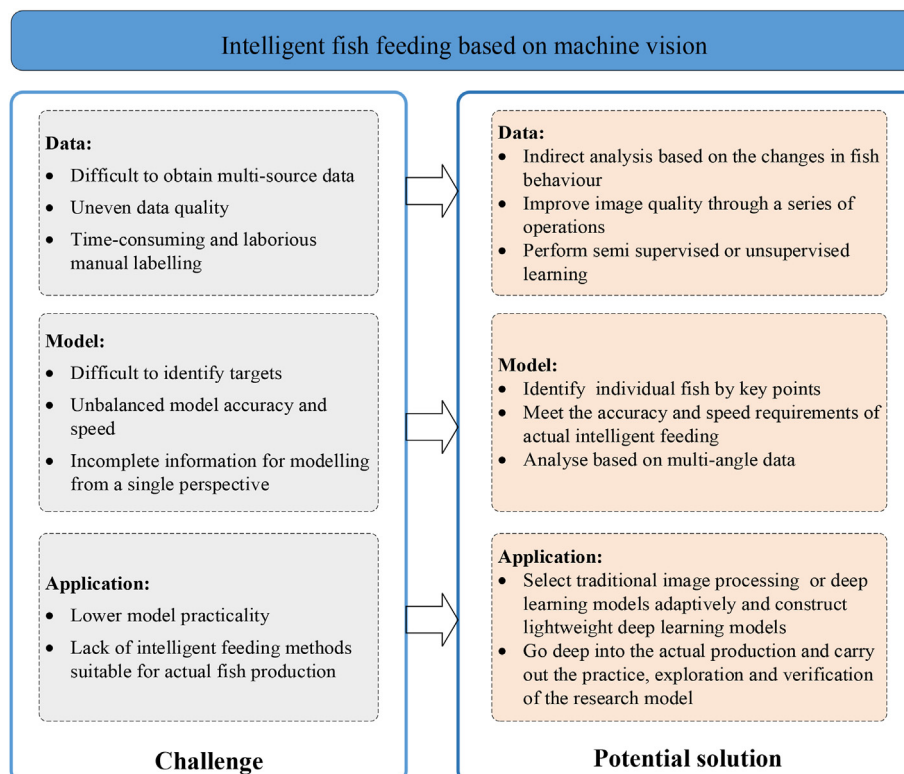


Fig. 15 – Challenges and potential solutions for intelligent fish feeding based on machine vision.

image is dense, the labelling effort is even greater. Therefore, how to get more information from unlabelled data and perform semi-supervised or unsupervised learning is a research direction in the future.

7.2. Model perspective

7.2.1. Difficult to identify target individuals

The fish image background is complex and diverse due to the different aquaculture environments and equipment configurations in different modes such as pond, industry, and cage. In addition, due to the high-density farming, gathering, chasing, and feeding behaviour of fish in the real scene, some phenomena, such as occlusion, overlapping, and deformation may occur between fish. Multiple overlapping targets can be easily identified as a single target, leading to inaccurate identification of fish individuals. Moreover, fish of different species and growth stages have different shapes and sizes, further complicating the identification of individual fish. Therefore, methods that identify key points such as fish eyes and fish heads help overcome the above problems. At the same time, considering all the targets in the image as a whole and taking the group as the target object to be analysed is another solution that is particularly effective in solving the problem of aquaculture water feed detection.

7.2.2. Unbalanced model accuracy and speed

The deep learning model has advantages in processing fish images with complex backgrounds and different fish scales. However, the deep network structure and a large amount of data make the training and inference time very long. In general, the more layers in the network, the higher the accuracy of the model, but the slower the speed. In response to this problem, we should analyse from the perspective of actual demand to understand the accuracy and speed requirements in fish intelligent feeding, and whether the cost of achieving higher precision and speed can save higher feed costs is also needed to know. To achieve the ultimate goal of maximising economic benefits, the accuracy and speed of the model should be reasonably balanced.

7.2.3. Incomplete information for modelling from a single perspective

The fish images collected from a single top view, side view, or underwater angle can only capture the variation of fish features in a particular projection plane, often resulting in a lack of information on fish depth and preventing a comprehensive analysis. Therefore, it is more reasonable to collect multi-angle data of fish based on multi-camera or binocular cameras, perform multi-angle data fusion analysis, extend from surface analysis to spatial analysis, and create a three-dimensional model of fish.

7.3. Application perspective

7.3.1. Lower model practicality

The traditional image processing model is simple, flexible, and fast. However, it is difficult to process images with complex and diverse backgrounds, which makes it difficult to meet the application needs of different breeding sites and modes in real

production. The deep learning model is not limited by the complexity and diversity of the images, but it requires a high level of computational power, and the hardware cost is high. In addition, the model size is generally large, making it difficult to carry on smart terminals such as mobile phones, and its practicability needs to be improved. Therefore, traditional image processing models or deep learning models are automatically selected for different research goals to reduce hardware costs based on meeting demand. In addition, lightweight deep learning models need to be investigated.

7.3.2. Lack of intelligent feeding methods suitable for the actual production of fish

Current research on intelligent feeding has failed to integrate the various influencing factors of fish appetite and to design a feeding method that dynamically adapts to the fish feeding demand. Further research is needed to provide specific parameters such as feeding amount, feeding frequency, and feeding time. Therefore, how to give specific feeding strategies and make a breakthrough from theoretical analysis to practical application is an important issue that needs to be addressed. To address this issue, we should go deep into the actual production and carry out practical practice, exploration, and verification.

8. Conclusion

This paper mainly reviews the research on fish intelligent feeding based on machine vision. The main factors affecting feeding are introduced, followed by a general description of the process of intelligent fish feeding based on machine vision, and a detailed overview and analysis of the key technologies involved in image acquisition, image processing and analysis, and intelligent fish feeding. Feeding challenges and potential solutions are also discussed. Compared with the existing feeding method, the fish feeding method based on machine vision has the characteristics of automatic, high efficiency, and non-destructive, which can improve the benefits and efficiency of aquaculture.

Nevertheless, there are still some shortcomings and challenges in the application of machine vision for intelligent fish feeding. The complexity and incompleteness of multi-source data acquisition and the uneven quality of the data itself make the source of the research a major obstacle; complex backgrounds and multi-scale fish size make it difficult to identify target individuals in the traditional image processing methods; and the need for large amounts of data for deep learning is also a problem that requires efforts to solve. In addition, research is needed on how to balance the model accuracy and speed, and how to mine fish information from multiple angles. At the same time, in terms of application, the practicability of the model should be further improved based on these shortcomings, targeted exploration, solution research, in-depth digging, and key breakthroughs.

With the development of the Internet of Things, big data, and artificial intelligence technologies, the application of machine vision in intelligent fish feeding has become possible. Future research should further solve the problems encountered, realise automatic feeding based on fish feeding demand, and apply the research models to actual production,

to promote accurate feeding in aquaculture and improve the efficiency of aquaculture.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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